

Where to Study or What to Study?
Changing Horizontal Stratification in Bachelor's Degrees in the
21st Century

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Abstract

A central question in the social stratification literature and for individuals interfacing with the US system of higher education is whether college graduates' earnings depend more on *what* they study or *where* they study. Drawing on a novel dataset linking US bachelor's graduates' specific educational credentials—defined as the combination of degree level, awarding institution, and field of study—to realized earnings and industry destinations from 2001 onward, this study provides the first comprehensive estimates of the changing roles of horizontal stratification across institutions and fields in structuring earnings inequality. Field of study predicts earnings more powerfully than institutional affiliation, and its influence has modestly increased over time. This growing salience reflects the intensifying alignment between specific fields and high-earnings sectors such as technology, finance, and professional services. Although variation across institutions accounts for a smaller share of inequality, it is increasingly correlated with characteristics such as selectivity and enrollment numbers. These changes are not driven by shifts in enrollment patterns, nor by demographic recomposition, and there is no evidence that high-earning credentials have become increasingly concentrated within high-earning institutions. These findings point to an ongoing reconfiguration of horizontal stratification in which field-specific ties to high-earnings industries and the evolving ecology of educational institutions shape earnings outcomes in new and consequential ways.

Introduction

The labor market returns to higher education are well documented (for reviews, see Card 1999; Hout 2012; Meghir and Rivkin 2011; Posselt and Grodsky 2017). Yet these returns vary substantially by institution (Zimmerman 2019; Brand 2006; Sekhri 2020), by field of study¹ (Altonji, Kahn, and Speer 2016; Kim, Tamborini, and Sakamoto 2015; Goyette and Mullen 2006; Kirkeboen, Leuven, and Mogstad 2016), and by their intersections, here termed specific education *credentials* to refer to the unique combination of degree level, awarding institution, and field of study.² However, set against the backdrop of the rapid expansion of higher education, technological change, and industry restructuring, it is unclear how returns to a degree are differentiated by *what* and *where* students study or how these categorical distinctions’ relative importance has changed across successive graduating cohorts. The present research is the first to systematically study both forms of *horizontal stratification*³, or the variation among otherwise equivalent vertically stratified degrees, in tandem and over time. Figure 1 provides a stylized representation of the forms of horizontal stratification discussed in this article as compared to vertical stratification.

In this paper, I show that institutional, field-based, and specific credential-based disparities together explain considerable variation in realized earnings among workers with at least a bachelor’s degree. However, the contributions of these factors to earnings are

¹I use the term “field of study” instead of “college major” to account for variation in degree types (e.g., B.A., B.S.N., B.F.A.) and to allow generalizability across vertical levels of education. “Academic program” might be a suitable alternative, though no term is fully satisfactory.

²While this notion of “credential” departs slightly from its typical usage (see Collins et al. 1979), it has precedent. I use credential to refer to specific institution- and field-specific degree designations, excluding post-degree licensure and other non-degree certifications. Alternatives like “departments” miss that the same department may issue different degree types (e.g., B.A., B.S.). As in Collins, credentials are labor-market currency—requirements regulating access to jobs and advanced study—though my scope is narrower and more granular. Here, the field of study- and institution-specific credentials here are nested within Collins’s broader concept, preserving its gatekeeping/allocative role while distinguishing, for example, the same degree across universities. This usage is also consistent with Brown (2001).

³I follow the stratification literature in defining horizontal stratification as variation within a given level of completed education. In this case, that is among individuals who have finished a bachelor’s

neither uniform nor static, showing distinct evolutionary trends. I find that field of study is a stronger predictor of earnings than institutional affiliation, and its relative importance is growing modestly across recent cohorts. The widening field of study differentials are driven primarily changing labor market dynamics, here proxied by flows from fields of study to specific industries. Field of study differences appear to be well explained by the increasing concentration of high earnings in a few industries to which certain fields of study are tightly linked. Meanwhile, institutional variation accounts for a smaller share of inequality, but it has become more structured by observable characteristics such as test scores and enrollment size. I find little evidence that high-earning fields of study have become more concentrated within high-earning institutions, that shifts in enrollments cause these patterns, or that changing demographics across fields of study offer a competing explanation for these findings. The changing importance of horizontal stratification has implications for theories of credentialism and social closure as well as models of status-attainment, recasting a bachelor's degree not as a unitary equalizer but as a constantly evolving engine of inequality in which differentiated educational credentials confer advantage among graduates.

The present research builds on prior studies that have typically examined each form of horizontal stratification in isolation and from cross-sectional, static perspectives. While prior work documents large and persistent pay gaps across fields and premiums associated with institutional characteristics (for reviews, see Gerber and Cheung 2008; Reimer and Thomsen 2019), analyzing each axis separately can conflate field-of-study and institutional effects because fields are unevenly distributed across institutions in path-dependent and sometimes idiosyncratic ways. Institutional and field of study differ-

degree. This usage differs from some strands of higher education research, where the term often refers to variation in experiences, resources, or outcomes during college, including between completers and non-completers. My focus is limited to degree recipients, which makes horizontal and vertical stratification analytically separable. Institutional and field of study-level differences in degree completion, being the result of forces that dictate vertical stratification, are important, but they are outside of the scope of this study.

ences also interact with the demographic characteristics of students in a given program, which is in turn shaped by changing regional population compositions and institution-specific student bodies, underscoring the need for institution-level analyses (Hamilton et al. 2024). Additionally, the predominance of cross-sectional designs obscures how the relative importance of institutions, fields, and their intersections has shifted across cohorts. Table 1 summarizes exemplary research by axis of horizontal stratification and time horizon. Along with Bleemer and Quincy (2025), the present study addresses a key gap by jointly analyzing both axes and tracing their effects across graduating cohorts. Whereas their study operationalizes horizontal stratification primarily as the heterogeneous effects of vertical stratification and relies on survey data that preclude direct analysis of horizontal stratification, I use fine-grained administrative data to trace how specific credentials themselves map to realized earnings and how those payoffs evolve over time.

The mechanisms underpinning the findings described in this paper are central to understanding how employers interpret educational signals in recruitment and job assignment, speaking directly to research on hiring biases and the social processes through which credentials acquire labor-market value. Although this study captures only part of the relational foundations of horizontal stratification, the inequalities documented here point to a broader research agenda tracing how educational credentials are embedded in employer networks and how those ties transmit value through concrete relational linkages. They also point to a clear invitation for expanded administrative data linkages and access to facilitate novel forms of inequality research. Taken together, the findings I document indicate an alternative lens through which one can view how higher education structures inequality. Where much stratification research focuses on access to and completion of college, the present study examines how distinctions among programs and institutions create unequal value within the bachelor’s degree. This shifts attention toward the organizational differences that prior work by Stevens et al. (2008)

highlights as central to understanding colleges as stratifying institutions and as hubs that connect students to the labor market and the larger economy. In doing so, the analysis links specific credentials to distinct opportunity structures and distributions of economic returns and points toward the political and institutional forces that give rise to these arrangements.

In the analyses that follow, to maintain analytical tractability I focus on a single level of vertical stratification: bachelor's degree holders. The choice is practical and theoretical as bachelor's degrees are the most commonly awarded U.S. postsecondary credential and a key gateway to many labor-market opportunities. Although often treated as a uniform marker of status, outcomes among recipients vary widely by institution and field, making this a critical site of horizontal differentiation. As the college-no-college earnings gap has stabilized (Autor et al. 2020) and the number of bachelor's degrees has nearly doubled since the turn of the century,⁴ new forms of inequality have likely emerged within higher education, organized less by vertical access than by the horizontal differentiation of programs and institutions whose ties to labor markets evolve alongside changes in industrial organization, wage-setting, and occupational closure. Further, the case of U.S. liberal arts system, where students can choose or change fields after enrolling and schools are rarely specialized, unlike more rigid systems such as Germany's, allows a clear test of whether institutional or field characteristics matter more for inequality. I track outcomes at one, five, and ten years after graduation to capture both the immediate effect of specific credentials and the indirect effects that operate through further study and job moves. I report five-year estimates in the main text, with other time horizons presented in the supplementary appendix.

*** Figure 1 About Here ***

⁴https://nces.ed.gov/programs/digest/d12_310.asp

Horizontal Stratification by Field of Study and Institution

Having motivated the need for the study of changing horizontal stratification in higher education over time, the next step is to clarify *how* horizontally stratified educational credentials map onto realized earnings in the labor market. Beginning with fields of study, Carnevale, Cheah, and Strohl (2013) show that earnings gaps across fields of study can exceed the overall college versus high school gap in earnings. In a human capital view, fields confer different skill bundles with different market premiums, which separates the *general* value of a bachelor’s degree from its *specific* forms (Kinsler and Pavan 2015; Kogan et al. 2021; Kambourov and Manovskii 2009; van de Werfhorst et al. 2001). Yet, it is not clear that field of study effects are reducible to skills alone. Rather than simply transmitting distinct bundles of human capital, some disciplines also function as prestige-laden credentials that signal valued capacities, channel students into occupationally closed or professionally regulated pathways, and attract pay premiums through mechanisms other than workers’ actual skills (Brown 2001; Thurow 1975). These processes situate educational credential hierarchies within status cultures and organizational practices that institutionalize prestige distinctions inside higher education. Fields of study may thus operate as status-bearing credentials as much as ascriptions of skill (see also Bills 2003).

Educational institutions themselves represent the other main axis of horizontal stratification, shaping outcomes through largely other mechanisms altogether. University prestige, for instance, affects returns to a bachelor’s degree net of ability or learning (MacLeod et al. 2017), cultural matching in elite labor markets privileges graduates of high-status universities (Rivera 2012), and even randomized exposure to elite peers can enhance long-run outcomes (Michelman, Price, and Zimmerman 2022). As with field of study, institutional premiums thus reflect a mixture of real and perceived advantages for individuals on the labor market.

Much of the prior literature estimates the effects of fields of study and institutions in isolation, yet in most survey data these forces are inherently intertwined because fields of study are unevenly distributed across institutions in unobserved but meaningful ways. For example, a measured premium from attending a flagship university may be real in a reduced form sense, but it can be driven by specialized programs offered only at such institutions, blurring whether the return should be attributed to the institution attended or the field studied. The data and methods I use allow these influences to be separated, but the goal is not to stage a contest of magnitudes for its own sake. Rather, since institutions and fields are analytically distinct sites where social and human capital are produced, organized, and valued, and they operate through related yet largely differentiated pathways, measuring their relative contributions clarifies where the main forms of labor-market advantage are generated and transmitted. Institutional prestige communicates social exclusivity, network quality, and a generalized status that travels with graduates across labor-market settings. Fields of study communicate domain-specific capital, including legitimate knowledge, cultural knowledge, occupational socialization, and, in some disciplines, specialized prestige signals that carry meaning within particular labor-market segments. These symbolic processes combine with genuine differences in training and skills that specific educational credentials confer. The contribution of this analysis is to measure how these two axes, and their intersections, jointly structure economic returns, while setting the stage for further work that traces the concrete mechanisms through which those returns are produced.

The aforementioned inseparability of field and institution effects inherent to survey-based research is further compounded by how both axes of horizontal stratification are captured in survey data. Institutions are proxied by observed attributes or collapsed into broad groupings (e.g., average test scores or selectivity tiers), while fields of study are recorded crudely and then aggregated into overly broad categories. These strategies obscure meaningful differences across institutions and flatten the closure-generating

processes that distinguish specific fields. For example, surveys often collapse computer science, information systems, and electrical engineering into a single “computer-related” major despite sharp differences in training, prestige, and labor market returns. Similar flattening occurs within medical and engineering programs, whose subfields have widely divergent outcomes but are too complex for many survey instruments. At the institutional level, treating all flagship publics as equivalent ignores sharp cross-state differences in resources, selectivity, and outcomes, just as collapsing private institutions into a single category erases the gulf between elite research universities (e.g., Ivy League schools) and tuition-dependent regional colleges. As for specific credentials, prior work rarely models fields of study within specific institutions directly. Where it does, evidence suggests limited institutional variation net of department effects in a Norwegian context (see Borgen and Mastekaasa 2018).

Correct measurement of degree characteristics is thus essential, as the value of fields of study and institutions reflects the specific human and social capitals they confer or signal and the labor-market destinations they open up. For instance, some fields of study place graduates in high-paying sectors with limited growth (for example, nursing), while others start lower but yield steeper long-run trajectories (for example, medicine). Still others, such as technology or consulting, combine high entry pay with sustained mobility (Cheng and Song 2019). These life-course patterns are further shaped by differences in graduate-school entry, and specific departments and the credentials they confer vary in their reputations and subsequent success in sending students onward to advanced degrees.

A central premise of this article is that such trajectories should be understood as properties of specific credentials and their institutional locations rather than as the simple aggregation of individual choices. The average work-related trajectories of students across different credentials and fields of study are also shaped by broader structural forces, including changes in occupations, industries, and forms of work. Many fields of

study are additionally embedded in national academic policies and vocational systems, highlighting the dialectical nature of the school-to-work frontier. International research on school-to-work linkages shows that stronger institutional linkages can reduce unemployment and improve job outcomes, although they may limit flexibility and increase mismatch (DiPrete et al. 2017; Bol et al. 2019). This literature, like the perspective advanced in the present paper, situates credentials within institutional structures. However, most prior work has been cross-national, whereas the present study examines how these relationships evolve over time within a single, large national context. This focus shifts attention from individual selection processes to institutionalized organizational and labor-market forces that shape credential outcomes and motivates the paper’s core strategy of tracing how links between credentials and segments of the labor market are reconfigured across graduating cohorts.

While the literature on horizontal stratification in higher education is rich in many respects, the aforementioned measurement difficulties inherent to survey-based research have led to research that has hitherto pursued a narrow agenda in other respects. Work on institutional effects in the United States has often focused on elite universities even though they educate a small share of students.⁵ However, most bachelor’s degrees are awarded by colleges that differ in enrollment patterns, residential versus commuter context, online share, and admissions and advising practices, which conventional metrics such as SAT scores or selectivity do not fully capture (Ciocca Eller 2023). Many students attend nearby colleges or face financial and family constraints, which makes variation among non-elite institutions central. In terms of intergenerational mobility, higher selectivity does not uniformly raise returns because many colleges with the highest mobility rates accept most applicants (Chetty et al. 2017). Likewise much of the literature on the effects of field of study differences focuses on the STEM–non-STEM dichotomy or on skill-biased technological change, leaving many coarse and fine-grained differences in

⁵By the author’s calculation, 21 Ivy Plus colleges account for 1.8% of bachelor’s recipients.

college majors unexamined.

Finally, beyond institutional selectivity or quality indicators, organizational features of universities and campus ecologies channel students unevenly by class background, producing divergent trajectories even within the same school (Armstrong and Hamilton 2013) and making an attention to specific institutions and credentials paramount. Crucially, outcomes depend both on where students enroll and on what they study there, because institutions differ in disciplinary strengths and students sort into programs in patterned ways. Among research-intensive elite universities, program mixes are often relatively similar, which can mask this interaction. Among the far larger set of non-elite colleges, the specific fields of studies offered at a given institution are of first order importance.⁶ Finally, not all educational credentials express the simple additive effects of a field of study and an institution, as departmental quality and program reputation vary. For these reasons, an analysis of inequality broadly amongst the full range of institutional flavors must jointly account for institutions and fields of study to capture how these environments together structure earnings.

Selection Effects and Measuring the Value Added of Horizontally Stratified Degrees

Observed differences in outcomes between horizontally stratified degrees does not necessarily reflect the quality or content of specific degrees. In the context of horizontal stratification, I use the term *selection effects* to refer to differences in outcomes across institutions, field of study, and their intersections that arise from pre-enrollment sorting. This usage is complementary to *value-added* or causal effects, which capture returns attributable to the training, networks, placement, and other features of the credential itself while holding student characteristics pre-enrollment constant. Distinguishing selection

⁶For example, many allied health professions (one of the largest groups of professions comprising laboratory technicians, respiratory therapists, dietitians, etc.) require specialty degrees only available at select universities in each state. Likewise, vocationalized bachelor's degrees for careers as accountants, commercial pilots, and police officers are often geographically limited within a given state.

effects from values-added is central because institutions and fields, and college itself, attract systematically different populations along dimensions correlated with earnings, including academic preparation, socioeconomic background, and labor-market aspirations.

Field of study and institutions differ in demographic makeup and in the forces that channel students into them, including preparation, aspirations, and perceived labor-market risk (Altonji, Blom, and Meghir 2012; Zhou 2019; Brewer, Eide, and Ehrenberg 1999; Loury and Garman 1995; Goyette and Mullen 2006). However, differential enrollment by demographic characteristics alone does not contaminate causal conclusions from observational data on earnings gaps. Problems arise when imbalances are *correlated with unobserved determinants of earnings* such as pre-college abilities, productivity, or motivation, or when the propensity to choose a specific field of study or college is correlated with the benefit a student would receive from a program (treatment effect heterogeneity bias; Zhou and Xie 2020). When such selection on unobservables is present, cross-credential gaps can reflect sorting rather than credential value added. Put differently, demographic segregation matters for the present study only insofar as it generates the earnings differences observed across credentials. Otherwise it is secondary to the task of describing shifting dynamics of earnings inequality over time, though it has implications for whom such inequalities affect the most. Evidence for selection effects driving institutional differences is mixed, with some studies attributing gaps to selectivity and others finding little effect once unobserved heterogeneity is addressed (Borgen 2014; Bleemer 2021; Manski and Wise 1983; Dale and Krueger 2002; Brand 2006; Gerber and Cheung 2008). By contrast, recent work provides limited support for the view that field-of-study pay gaps primarily reflect selection (Dahl et al. 2023; Bleemer and Mehta 2022; Kim et al. 2015).

A recent comprehensive study of the college mobility pipeline by Bleemer and Quincy (2025) puts selectivity in perspective with regards to the present study. Drawing on more

than a century of survey data, the authors show that college has become increasingly regressive since 1960 (the benefits of vertical stratification have shifted toward higher-income groups) due to structural changes in higher education. In doing so, the study also quantifies how much between-institution and between-field of study earnings differences reflect the value added of a degree (the value net of selection), finding that selection explains only a small share. After accounting for selection effects, about 70–80% of institutional differences and nearly 100% of field differences reflect causal effects of credentials rather than preexisting student differences (Bleemer and Quincy 2025). The institutional estimate is consistent with Chetty, Deming, and Friedman (2023).

It is intuitive that selection into institutions and fields explains only a limited share of horizontal inequality. Along the vertical margin, such as college versus high school, selection is strong because groups differ in preparation, resources, and aspirations, all of which correlate with earnings even after accounting for college attendance. Along the horizontal margin, differences across schools, fields, and their intersections arise from more stochastic forces that do not reliably track prior ability in ways tied to potential earnings. Xie et al. (2015) show this for STEM fields of study: family background and individual traits matter for college preparation, yet they do not predict choosing a STEM field. Students more often land in programs through advising nudges, personal interests, proximity and commuting constraints, tuition and net price, and frictions such as course availability, lotteries, and enrollment priorities. These processes sort students in ways that are noisy with respect to ability, so observed gaps across credentials largely reflect credential value added and the labor-market destinations those credentials enable (Bleemer and Quincy 2025). Demographic sorting still matters for who gains entry, but the earnings attached to credentials appear more sensitive to processes that occur after students sort into programs. This is consistent with the literature on horizontal stratification and its links to demographic inequality. Hamilton et al. (2024) document how horizontal stratification coincides with racial, class, and gender inequality in access.

Their perspective clarifies who gets in, while my analysis focuses on how the value of credentials shifts and how those shifts translate into labor-market outcomes, which helps explain subsequent stratification across demographic groups.

My analysis is possible even though I do not model selection directly. The findings in Bleemer and Quincy (2025) offer a useful benchmark and help situate my results within broader patterns of inequality. My contribution is to center the structure and evolution of horizontal stratification itself, asking how the value of credentials has shifted over time and how these shifts reflect changes in higher education and the labor market. Unlike survey designs, the dataset I use permits fine-grained analysis at the credential level and makes it possible to separate field effects from institutional effects while tracking their joint evolution. Nevertheless, to address concerns that composition alone drives the results, I implement several robustness checks, described below. I now turn to the analytical overview, where I lay out the measurement strategies I use to quantify horizontal stratification across fields of study, institutions, and their intersections and to identify the forces undergirding changes within each stratifying domain.

*** Table 1 About Here ***

Analytical Overview

This paper treats field of study, institution, and their intersection as analytically distinct yet interdependent dimensions of horizontal stratification. Each represents a categorical form of differentiation within a single level of education but operates through different underlying mechanisms. Together, they constitute the primary ways in which educational differentiation structures labor market outcomes among bachelor’s degree recipients.

Fields of study shape earnings primarily through differential flows to labor-market destinations that vary in work content. Their effects arise from how programs channel

graduates into specific industries, firms, and occupations, which are themselves characterized by distinct wage-setting regimes, internal labor markets, and opportunities for advancement (DiPrete et al. 2017; Bol et al. 2019; Kalleberg and Lincoln 1988; Tomaskovic-Devey et al. 2020). As between-industry and between-firm dispersion in wages has grown (Song et al. 2019; Wilmers and Aeppli 2021; Haltiwanger et al. 2024), the labor market destinations associated with a field have potentially become an increasingly consequential determinant of its average returns. Therefore, the analyses that follow seek to disentangle the extent to which horizontal stratification by fields of study is due to changes in fields of study, changes in labor markets, and changes in how they are linked to one another.

Institutions, in contrast, influence labor market outcomes largely through prestige signaling, resource endowments, and social capital formation. Institutional prestige conveys general status and exclusivity that may operate independently of any particular field of study, and institutional environments vary in peer composition, advising quality, and access to employer networks, all of which shape post-graduation opportunities (Rivera 2012; MacLeod et al. 2017; Michelman et al. 2022; Ciocca Eller 2023; Armstrong and Hamilton 2013). While selectivity and observable characteristics such as tuition or test scores proxy parts of these mechanisms, a substantial portion of institutional value likely remains embedded in organizational practices and campus cultures that produce portable but often intangible advantages. A key task of this analysis is to measure how much of variation in institutional outcomes is due to observable differences in institutions as opposed to unobservable differences in institutions.

The intersections of field and institution, institution- and field of study-specific educational credentials, combine these mechanisms in distinctive ways. Certain departments at particular universities possess well-developed linkages to specific employers or industries, creating pipelines that cannot be explained by either institutional prestige or field-level skill specificity alone. Such credential-level linkages can consolidate new forms

of inequality by directing graduates toward high- or low-earnings sectors in systematically different ways. For this reason, I take the institution-by-field of study combination as an operative unit alongside its margins. In addition to measuring how such combinations systematically dictate earnings in college graduates, I also seek to understand how fields of study are differentially distributed across institutions in patterned ways to see if certain colleges and universities have been better able to adapt their course offerings in a changing labor market.

The empirical design I employ below follows from these mechanisms. In the first part of the analysis, I reconstruct individual-level earnings from aggregated Post-Secondary Employment Outcomes (PSEO) data containing field of study-by-institution, specific credential-level statistics to recover within-credential dispersion and decompose total earnings inequality across cohorts into components attributable to institutions, fields of study, and their intersections. This approach also distinguishes changes arising from shifts in credential-specific earnings premiums from those due changes in overall enrollment patterns. In the second part, I disentangle each major axis of horizontal stratification independently. For fields of study, I analyze changes in average earnings through field-to-industry flows and the earnings structures of destination industries, decomposing whether trends are driven by reallocation across industries or by changes in industry pay. I also benchmark these changes against what would be expected from demographic recomposition within fields. For institutions, I assess the extent to which observable characteristics such as selectivity, size, tuition, and enrollment explain variation in outcomes net of field composition and how these associations evolve over time. For credentials, I examine whether field-institution pairings have become more stratified—specifically, whether high-earning fields have become increasingly concentrated in high-earning institutions—thereby indicating consolidation of advantage at the intersection of prestige and field-specific specialization.

Data

Across both parts of the analysis, this study draws primarily on the U.S. Census Bureau’s experimental PSEO dataset, which links college graduation records to employer-reported earnings via the Longitudinal Employer–Household Dynamics (LEHD) system. To my knowledge, this study represents the first use of these data in sociology research. The LEHD provides near-complete coverage of annual earnings reported for unemployment insurance purposes in participating states, and the PSEO links these earnings to graduates by institution, field of study (four-digit CIP code), and graduation year. The result is a dataset that captures labor market outcomes at the credential level for up to ten years after graduation. This structure enables direct measurement of horizontal stratification along the dimensions outlined in the introduction.

The PSEO contains two core components. The “earnings” dataset reports average earnings for each credential one, five, and ten years after graduation, alongside the 25th and 75th percentiles of the earnings distribution. The “flows” dataset tracks the number of graduates from each credential entering different industries, aggregated at the two-digit NAICS level, at the same intervals. The data also include counts at the credential level of workers not employed in the workforce, based on the observed gap between the number of graduates and the number of workers found in the LEHD. Although the data are aggregated, they allow partial reconstruction of earnings distributions and analysis of how field–industry linkages evolve over time, which I discuss below. I focus on five-year post-graduation outcomes, which reflect earnings after one has established their career and avoid the limited cohort coverage in the ten-year data. Five-year post-graduation data includes earnings even for individuals who have continued their education, so long as they are in the workforce at the time of follow-up. Otherwise put, the effects shown here represent the *reduced form effect* of a bachelor’s degree or the total effect of a bachelor’s degree, even if it is mediated by continued education within five years of graduation.

While a number of students may not have finished their schooling five years after they graduate with a bachelor’s degree, most have if they are going to. For comparison’s sake, one- and ten-year outcomes are used in supplementary analyses that double as robustness checks to the extent that these results are mediated by continued graduate education. Because of this decision, results in the main analyses are only available for cohorts up to 2013–2015, as they must be observed five years after graduation, and the raw underlying data extend through 2019. More recent results are available in the appendix for a one year time horizon.

Because the PSEO is based on voluntary state and institution agreements, its coverage is incomplete but diverse, spanning public and private universities, community colleges, online colleges, and flagship state institutions. While not nationally representative on their own, the breadth of institutions allows for rich analyses of variation in credential outcomes. The fact that many states’ participating institutions represent over 80% of these states’ total graduates allows the sample to credibly reflect a representative sample nationwide of college graduates after reweighting.⁷ I adjust for imbalances in institutional representation in the original data using an entropy balancing reweighting method discussed in the supplementary appendix, and I revisit these limitations later in the paper. Appendix Figure A1 maps institutional coverage.

One important limitation is that the PSEO does not report earnings by industry at the credential level. While I observe the average earnings for each credential and the distribution of its graduates across industries, I cannot observe earnings within industries conditional on credential. To approximate this, I supplement with industry-level earnings data from the American Community Survey (ACS), which reports average annual earnings by industry for bachelor’s degree holders. These estimates are used only in analyses where industry flows and sectoral average earnings are central, which are just a subset of all analyses. The PSEO data also do not disaggregate credential-level earnings

⁷https://lehd.ces.census.gov/data/pseo_experimental.html

by demographic characteristics: race, ethnicity, and gender. I undertake a sensitivity analysis using NCES data, which contain cohort-by-cohort credential-level statistics of degrees conferred by race/ethnicity and gender.

Finally, I incorporate data from the Department of Education’s College Scorecard to characterize institutional attributes such as mean SAT scores, tuition, undergraduate enrollment, and selectivity. These variables are used to evaluate whether observable institutional traits help explain differences in graduate earnings net of field composition. They are also used in the reweighting of the data to ensure representativeness. As with the ACS data described in the previous paragraph, these data are used in supplementary analyses that examine institutional-level variation and are not central to the paper’s headline findings.

Part I - Decomposing Variation in Earnings Inequality Amongst College Graduates

Methods

Generating Income Distributions From Observed and Scenario-Based Credential-Level Earnings Summary Statistics

I begin by simulating individual-level income data from specific educational credential-specific and cohort-specific earnings distributions. This allows me to contextualize categorical distinctions within the broader earnings distribution and compare inequality between credentials to inequality among individuals within the same credential. To do so, I simulate individual-level income data from the aggregate statistics provided in the PSEO dataset.⁸ This allows the importance of between-field, -institution, and -credential variation to be compared relative to variation within these groupings.

⁸While a parametric variance decomposition is possible when distributions are presumed to be log-normal, it provides similar results to the approach taken here and is less flexible than a simulation-based approach. The simulation-based approach allows for flexibility in specifying the robustness tests, in particular varying distributional forms and adding in non-earners, as specified later in the methods section.

In the earnings dataset, for each specific credential (field of study-by-school combination) and graduation cohort combination, four statistical moments are known: the number of graduates, and the 25th, 50th, and 75th earnings percentiles (π). Graduation cohorts are grouped into three-year spans for bachelor’s degrees (e.g. 2001–2003, 2004–2006, 2007–2009, etc.). This aggregation is necessary to protect individual privacy as many credentials only graduate a few individuals each year, and sample sizes for each cohort-credential observation must be large enough to protect graduates’ privacy. On the other hand, this allows the inclusion of even small fields of study.⁹ Using these points of information, one can recover a simulation of the original microdata that produced them. To do so, I assume that within each cell (specific educational credential–graduation cohort combination), incomes are distributed log-normally. For each cell, the density function of the distribution is calculated as is shown below in equation 1, which calculates a singular standard deviation (equation 3) from the 25th and 75th percentiles. As incomes are assumed to be log normal, the logged median and logged mean are assumed to be the same (equation 2).

$$f(x) \sim \mathcal{N}(\mu, \sigma^2) \sim \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x - \ln(\mu))^2}{2\sigma^2}} \quad (1)$$

$$\mu = \ln(\pi_{50}(X)) \quad (2)$$

$$\sigma = \frac{\ln(\pi_{75}(X)) - \ln(\pi_{25}(X))}{2 \times 0.674} \quad (3)$$

Using the above functional forms, I simulate individual earnings from the reconstructed distribution for each credential, generating a dataset that approximates the original underlying microdata. This simulated dataset serves as the basis for most subsequent analyses. The method carries limitations: most notably, it assumes log-normality and no skew within each distribution. While deviations from log-normality could affect the tail behavior, this assumption is well supported in the literature (Gibrat 1931;

⁹Noise is also added to the data to keep them differentially private, though it is not expected that this process will substantively affect my results.

Battistin, Blundell, and Lewbel 2009). To ensure the robustness of this assumption, I implement an alternative specification using a piecewise log-normal/Pareto distribution, assigning a Pareto tail to the top 2% of the distribution, capturing greater income inequality among high earners in line with previous work. These approaches assume that individuals within each credential are interchangeable. Given that my analysis relies exclusively on aggregated data and not individual characteristics, this assumption is analytically appropriate.

In addition to simulating observed distributions, I generate counterfactual, scenario-based datasets that allow me to isolate the drivers of inequality over time. In creating simulated datasets of individuals for each credential, I hold constant at the earliest observation either the number of individuals in each credential or the credential-specific distribution of income across cohorts in what amounts to effectively a “Das Gupta” decomposition (1993). Using these simulated scenario-based datasets in the following sections allows me to determine whether it is shifting allocation of individuals across institutions, fields of study, and specific credentials that may be driving changes in stratification or if it is simply different average earnings attached to each that are driving the effects.

Variance Decomposition

With these simulated individual-level earnings data in hand, I next decompose the total variance in earnings to estimate how much is attributable to different forms of horizontal stratification: field of study, institution, and specific credentials. This variance decomposition approach allows me to assess the relative importance of each component and track how their contributions to inequality evolve across cohorts. I perform a variance decomposition of logged annual earnings at the individual level $\ln(\omega_{i,u,f}|Y)$ wherein I, separately for each graduation cohort (y) and year of followup (X), iteratively add fixed effects for field of study (F_f), institution/university (U_u), and institution-field of study

interactions $\left(\gamma(U_u \times F_f)\right)$. The differential increase in R^2 is the percent of the total variance explained by each additional term with respect to all previous terms. In comparing these estimates over time and across cohorts, one can recover the extent to which each axis of horizontal stratification matters among degree holders in determining annual earnings. The data used are reweighted to be more representative of all US colleges using methods discussed in the supplementary appendix. The model specification is as follows, and fixed effects components are added step-wise from left to right in equation 4¹⁰:

$$\ln(\omega_{i,u,f}|Y = y + X) = F_{f,y} + U_{u,y} + \gamma(U_{u,y} \times F_{f,y}) + \epsilon_{i,y} \quad (4)$$

While the order of the components will affect the results, particularly given that fields of study are unevenly distributed across universities and change over time, I address this concern through a sensitivity analysis in the supplementary appendix by reversing the order in which fixed effects are entered. Alternative approaches to this issue do exist, though they are generally designed either for settings with more than two primary fixed effects or for contexts where the primary object of interest is a single “main” regression coefficient (see, for example, Xie and Zhou 2014; Gelbach 2016). In contrast, my approach provides a transparent set of bounds: one in which field of study premiums precede institutional premiums in importance, and one in which the reverse is true. I present the former in the main text, while also discussing the latter in the results section and presenting its results in the supplementary appendix. However, the ordering I have chosen for the main results is not arbitrary; it reflects the reasonable view that institutional earnings premiums are layered on top of, rather than independent from, field of study premiums, such that an institutional premium is meaningfully interpreted

¹⁰The estimated coefficients of the full model with field of study, institution, and their intersection fixed effects is not identifiable because the interaction term is nested within its main effects. This does not pose a problem for the present analysis because the variance of each component and its contribution to total explained variance are identified, however. The decomposition relies on the incremental change in R^2 as each set of fixed effects is added rather than on the absolute value of the coefficients themselves.

in relation to a given field of study premium. Otherwise put, a credential is associated with three additive premiums: one based on field of study and the skills and capital conferred with it, one based on institutional prestige and the various forms of capital that come with it (net of field of study), and one based on specific departmental- or credential-level traits (net of the prior two premiums).

*** Figure 2 About Here ***

Results

Data Coverage, Representativeness, and Descriptive Statistics

Figure A1 shows the geographic distribution of degree-granting institutions in the PSEO data. Inclusion is not based on a probabilistic sample but on voluntary partnerships between states, institutions, and the federal government. Figure 2 further illustrates this selectivity by plotting all U.S. universities by 2016 undergraduate enrollment, with in-state tuition, average SAT scores, and admission rates where available. Included and omitted schools are distinguished by color. While many private universities are missing, this is not a major concern: within public and private sectors, included schools largely mirror the full population in their institutional characteristics, though coverage varies. In this sense, the data are structurally representative, though compositional reweighting is necessary. One exception is the underrepresentation of Ivy League and similarly selective private colleges, which appear as outliers in the lower portions of the facets on selectivity and size. Given their small enrollments relative to large public institutions, their exclusion likely has limited effect on overall trends, though it constrains generalizability at the very top of the institutional earnings hierarchy, as discussed in the discussion.

*** Table 2 About Here ***

To make the data more closely reflect the actual target population of all U.S. institutions granting bachelor’s degrees, I reweight the included PSEO institutions using an entropy balancing technique so that, on observables, the weighted PSEO sample more closely matches the full universe of colleges and universities. Specifically, drawing covariates from the College Scorecard (public/private, enrollment, tuition, selectivity, and average SAT scores), I calculate weights to match first moments of these covariates in the target population while minimizing divergence from uniform weights. I apply the resulting weights in all analyses in this paper where appropriate. Covariate balance before/after weighting is shown in Table 2. Balance is largely achieved, but there is a slight mismatch in in-state tuition due to missingness of the most prestigious private colleges. Though highly selective publics like the University of Michigan are present in the data, these have lower tuition rates, making achieving perfect balance infeasible. For full details of these methods, see the supplementary appendix. To qualify and foreshadow further results below, many results are not changed dramatically by this reweighting exercise, and so imperfect balance is not much cause for alarm. This likely reflects three stylized facts: 1) Most students attend public schools (which are well represented in the PSEO data); 2) Even when they attend private schools, they are non-elite private schools (which are well represented in the PSEO data); and most importantly 3) Much meaningful variation between schools transcends typical metrics of school prestige and selectivity and comes down to more nuanced variation based on instruction type, resources, school environment, and advising structures, which makes rebalancing the data on these characteristics not have much impact. I provide direct evidence of the last point in Part II of the analysis.

*** Table 3 About Here ***

Table 3 shows the number of included units in the data, stratified by cohort. Because the PSEO aggregates credential-cohort combinations into multi-year credential-

cohort combinations, the data are reported in three-year spans for bachelor’s degrees (2001–2003, 2004–2006, etc.). In 2014, roughly 1.8 million individuals earned a bachelor’s degree in the United States. By comparison, my cumulative sample size of earnings five years post-graduation for the 2013–2015 graduating cohorts is 1,309,057, which represents slightly less than one-quarter of all U.S. bachelor’s recipients. This shortfall reflects both the non-coverage of certain institutions (as discussed above) and the fact that not all graduates are observed with earnings in the LEHD system. The PSEO does not track unemployment directly, but it does report the gap between the number of degrees conferred and the number of graduates with matched earnings records. This discrepancy reflects all individuals not linked to the LEHD as employees—a group that may include the unemployed, those out of the labor force (e.g., those engaged in full-time caregiving, pursuing further schooling, or working in informal work or work abroad), and those working in uncovered sectors. Because these individuals have no observed earnings, the analyses in the main text implicitly condition on labor-force attachment. As a result, my main estimates potentially mismeasure total inequality across all graduates if unemployment is considered. I account for this by simulating non-attached workers, as reported by the PSEO data for each specific credential-cohort combination, as having zero annual earnings in a sensitivity analysis, even though it is likely that many non-workers are not employed by choice. These results are presented in supplementary appendix Figure [A4](#) and are discussed where appropriate below.

In Table [3](#), it is clear that the number of individuals, fields of study, schools, and their intersections are increasing over the time period shown in the data, however colleges also begin being represented in the data at different points in time. To accommodate this, most of my analyses use a restricted set of institutions that appear in the data every year, amounting to 270 degree-granting institutions in total. In the restricted set, the number of college graduates increases by over 50% in the observed time period, which is commensurate with the growth of college attendance and population growth in

this time period. As the number of fields of study at the four-digit level is increasing over time, I also conduct sensitivity analyses using two-digit CIP codes, the number of which does not increase similarly over the period in question, and I use a middle ground between two- and four-digit CIP codes, which is detailed in the supplementary appendix. Furthermore, the Das Gupta-inspired analysis holds constant enrollments, allowing me to ensure that the creation and elimination of fields of study within and across institutions are not driving the results.

*** Table 4 About Here ***

*** Figure 3 About Here ***

Finally, to orient the reader to the structure of the original data before proceeding to my analytical results, Table 4 shows the top and bottom ten credentials in terms of earnings five years post-graduation for the bachelor's degrees graduating cohorts 2001–2003 and 2013–2015, the earliest and most recent cohorts of the sample. Even from this simple summary table alone, several notable phenomena stand out. For both sets of cohorts, the level of structured horizontal stratification is staggering. The lowest-earning credentials earn less than one-fifth of what the highest earning credentials earn. While the interquartile range of pay within each credential is large, it appears dwarfed by this between-credential variability in earnings. For earlier cohorts, engineering and pharmacy majors monopolized the highest earning credentials, while in later time periods it is dominated by a mixture of computer science, health professions, engineering and pharmacy. Also, the highest earning credentials appear to be much larger than the highest earning credentials in the earlier time period, with larger cohort sizes. Finally, the righthand-most column shows the percent of workers in the largest industry as a function of each credential. This is a simplified way of rendering the industry share distribution for each credential since there are 20 possible industries, but it shows that

the highest earning credentials appear to have stronger credential-to-work linkages than the lowest-earning credentials. Figure 3 shows this variation in an alternative manner across the entire distribution of credentials, displaying each credential as a point on the graphs, stratified by the parent-level two-digit field of study.

*** Figure 4 About Here ***

Earnings Variance Decomposition Analysis

Results for the variance decomposition are shown in the left-most panel of Figure 4. Over the period studied, the share of variance in log earnings explained by degree characteristics increased from about 25 to 30 percent five years after graduation. In other words, earnings among bachelor’s recipients have become more structured by observable features of their degrees and less by within-credential individual differences.

Several striking findings emerge. First, most of the variation in earnings is explained by what people study—not where they study. The magnitude is notable: for the 2013–2015 cohorts, 21% of variation is attributable to field of study, compared to just 5% for institution, and 4% for specific credentials. Second, the rising importance of degree characteristics is primarily driven by the growing influence of field of study. There is also slight evidence of increasing institutional importance, though to a much smaller extent. Third, the contribution of specific credentials has remained largely stable over time. Otherwise put, changing characteristics of individual departments, net of school and field effects, are not driving the results. These findings are robust to multiple sensitivity tests. Appendix Figure A4 shows consistent results across alternative specifications: using two-digit CIP codes, including non-earners as zeroes, applying a piecewise log-normal/Pareto distribution, estimating models without log transformation, estimating unweighted models, and using different weights to upweight only private institutions to match the national bachelor’s degree-holding population. Across all approaches, field of study remains the dominant and increasingly important predictor of earnings. I draw the

reader’s attention specifically to the robustness check that reorders the fixed effects to allow institutions to come before fields of study. Even here, where the institutional effect is measured first, allowing it to include the effect caused by institution-level differences in fields of study offered, the explanatory power of field of study prevails. Indeed, it is impossible to fully disentangle the two, but the main effect and this robustness check provide the lower and upper bounds of the importance of field of study and institution, with both scenarios highlighting the explanatory power of field of study.

*** Table 5 About Here ***

A natural follow-up to the previous results is whether observed changes are driven by changing enrollments across credentials or by shifts in credential-specific average annual earnings. To address this, I decompose the results from the left-most panel of Figure 4 using two counterfactual scenarios: one holding credential-specific earnings constant at their earliest observation, and another holding constant the size of graduating cohorts by credential. This Das Gupta-style decomposition shows that most of the change is due to shifting average earnings across credentials, not changing enrollments. In fact, such allocative forces appear to slightly offset the earnings-based trend across the first four cohorts observed. That is, students have not disproportionately moved into high- or low-earning credentials in ways that would explain the increasing inequality, nor has the creation or disappearance of specific credentials driven the results. Instead, the credentials themselves have changed in what they yield in the labor market. The causes of these shifts are unclear but are taken up in later sections. Table 5 shows that the flight from high-earning fields of study like business and computer science (discussed further below) is likely partially to blame for this. Appendix Figure A2 replicates the decomposition at both one and ten years post-graduation, with mostly consistent findings at both intervals. The trend of growing explanatory power, as driven by fields of study, also holds for one-year outcomes. While only three cohorts allow

ten-year follow-up, these results show no deviations that would call earlier findings into question, and the lack of inconsistent findings between the three different periods of follow-up indicate that the primary findings are likely not entirely mediated by graduate schooling.

While these results do not necessarily reflect the causal “premium” offered by specific fields of study, institutions, and credentials, or the “value added” of choosing one credential over another, prior studies of the causal effects of institutions and fields of study can benchmark the findings reported here. As mentioned in the introduction, Bleemer and Quincy (2025) synthesized the evidence on this subject and found that earnings differences in fields of study reflect 100% of the value added of attending different majors. For institutions, earnings differences may be attenuated by 20% when accounting for selection effects. Otherwise put, accounting for differential selection into fields of studies and institutions only would attenuate the findings reported here for institutions, and the importance of fields of study would remain, strengthening my main finding of the increased importance of field of study. There are no causal studies at the credential level on earnings differences, but they are likely somewhere between fields of study and institutions in representing the “value added” effect of credentials.

Finally, one might worry that the results are mechanistic as there are many more fields of study than one might expect. Even when repeating the analysis at the 2 digit CIP code level, which juxtaposes 36 fields of study against 270 institutions, fields of study retain their explanatory power (supplementary appendix Figure A4), though its power is slightly attenuated.

*** Figure 5 About Here ***

Part II - Understanding the Drivers of Each Axis of Horizontal Stratification

The methods and results described up to this point have enabled me to paint an overview of the relative importance of each kind of horizontal stratification in driving earnings inequality among college graduates over time. However, the fundamental forces driving these changes require further study. The following methods will tackle each source of horizontal stratification—field of study, degree-granting institution, and specific credentials—one at a time to understand the underlying forces dictating the importance of each.

Methods

Analyzing Changing Average Annual Earnings by Field of Study Over Time

To understand how and why field of study-based inequality in earnings is changing, I focus on the evolving relationship between fields of study and labor market destinations. I analyze flows from fields of study into specific industries, together with changes in industry-level earnings for college graduates. Because the PSEO data do not report earnings by industry, I combine them with ACS data on industry-level earnings among college-educated 27–29-year-olds, along with average earnings by field and observed field-to-industry flows. Under an assumption of additivity in logged components, this linkage allows me to estimate average earnings by field net of industry, and by industry net of field. Further methodological detail on the construction of these estimates is provided in the supplementary appendix.

These estimates may then be used to estimate expected field of study average earnings based on changing industry flows. Moving beyond such simple comparisons and to formally disentangle the contributions of changing industry composition and changing industry-specific average earnings to field of study-level earnings trends, I decompose

changes in $\ln(\omega_{f,y})$, logged average earnings for each graduating cohort in a given field of study, using an extended version of a Kitagawa-Oaxaca-Blinder Decomposition (Oaxaca and Sierminska 2023; Kitagawa 1955). The following equation (equation 5) describes a basic Kitagawa-Oaxaca-Blinder Decomposition using two time points, y_1 and y_2 , which are equivalent to the earliest and most recent cohorts in my sample. Average shares and average annual earnings (equations 6 and 7, respectively) are based on the simple means of the two time points. There is also an residual component reflecting the difference in the observed change in field of study-specific average annual earnings and what is predicted by industry shifts and changing industry premiums alone (equation 8).

$$\Delta \ln(\omega'_f) = \sum_{ind} \overbrace{S_{ind,f}^* \left(\ln(\omega_{ind,f,y_2}) - \ln(\omega_{ind,f,y_1}) \right)}^{\Delta \text{ due to changing industry annual earnings}} + \overbrace{\ln(\omega_{ind,f}^*) \left(S_{ind,f,y_2} - S_{ind,f,y_1} \right)}^{\Delta \text{ due to changing industry shares}} \quad (5)$$

$$S_{ind,f}^* = \frac{S_{ind,f,y_1} + S_{ind,f,y_2}}{2} \quad (6)$$

$$\ln(\omega_{ind,f}^*) = \frac{\ln(\omega_{ind,f,y_1}) + \ln(\omega_{ind,f,y_2})}{2} \quad (7)$$

$$\Delta \ln(\omega_f) = \Delta \ln(\omega'_f) + \varepsilon_f \quad (8)$$

Holding shares at their average amount between the two time points $S_{ind,f}^*$ and annual earnings at the average level between the two time points $\ln(\omega_{ind,f}^*)$, one can decompose the extent to which overall changes in average annual earnings by field of study are due to each component. However, since industry changes are affecting the entire labor market to some extent, I further decompose the change due to industry shares to a “global” and a “local” effect, where the former is the shift due to industry growth in the overall labor market of college graduates, and the latter is any industry shifts net of that.

Expanding the second term in equation 5, I arrive at the following expression:

$$\ln(\omega_{ind,f}^*) \left(\overbrace{(S_{ind,y_2} - S_{ind,y_1})}^{\text{Global effect}} + \overbrace{[(S_{ind,f,y_2} - S_{ind,f,y_1}) - (S_{ind,y_2} - S_{ind,y_1})]}^{\text{Local effect}} \right) \quad (9)$$

In sum, the above “extended” Kitagawa-Oaxaca-Blinder Decomposition allows me to decompose the extent to which changes in observed field of study average annual earnings are due to three components: global growth and decreases in industry representation of working college graduates, local field of study-specific growth and decreases in industry flows, and changes to average industry-specific average annual earnings among college graduates. Any residual differences (ε_f) between observed and predicted changes in annual earnings over the time period in question are presumed to be due to interactive effects between fields of study and industry, within-field heterogeneity in four-digit CIP codes since CIP codes are measured at the two-digit level, or due to individual variation. None of these fine-grained sources of variation can be modeled directly using the data, but the magnitude of the residual changes in total are shown for comparison.

Just because industry flows may predict field of study-level earnings, does not mean they are the primary cause. While this research is in not causal, it is important to acknowledge that changing field of study average earnings may simply reflect demographic compositional change within fields of study. Even if this were the case, it would not invalidate these methods and any findings that come about because of them, but it would be a prudent way to qualify my results. In order to test this possibility, I devise a simple test to predict cohort-to-cohort changes in average earnings based on within-field of study demographic composition as opposed to industry flows. I employ within-field of study fixed effects models wherein the predictors are alternatively demographic composition (which is available at the field of study and credential level in the NCES data) or industry flows and use the within-field of study predictive power to understand how well each phenomenon describes changing earnings. I present these robustness checks

as additional methods and results in the supplementary appendix, and I discuss these results in the main paper’s results section below. I also directly compare predicted earnings changes based on each method to realized changes in earnings in the supplementary appendix.

Analyzing Changing Average Annual Earnings by Degree-Granting Institution Over Time

Because institutional stratification operates through mechanisms largely distinct from those of field of study, such as prestige, teaching quality, resources, and student composition, it requires a different analytical approach. Net of the distribution of fields of study within a degree-granting institution, there are several variables along which institutions may be stratified: commuter vs. non-commuter schools, flagship vs. other state universities, private vs. public universities, average SAT score, tuition fees, and more. To understand changes in institution-specific average earnings over time, I assess whether observable characteristics, such as selectivity or size, predict institutional differences in earnings, net of field of study, and I assess these changing relationships over time.

With College Scorecard data, I use average SAT scores at the institution level, logged undergraduate tuition, the logged number of graduate students, the logged number of undergraduate students, and the undergraduate admission rate as indicators of college quality and prestige. As all variables are continuous in nature, they likely explain more variation in the data than do binary or categorical indicators representing institutional characteristics. To assess the degree to which they are associated with earnings after graduation, I regress them on adjusted average annual earnings for each university in a given year, $\Omega_{u,y}$, which are normalized for the distribution of fields of study within that institution. Otherwise put, $\Omega_{u,y}$ represents the average earnings of all graduates from a given university, adjusting for field of study level differences in earnings. It is calculated

as follows¹¹:

$$\Omega_{u,y} = \sum_f \frac{N_{f,u,y}}{N_{u,y}} (\bar{\omega}_{f,u,y} - \bar{\omega}_{f,y}) \quad (10)$$

Estimates of $\Omega_{u,y}$ are then regressed separately against each institutional characteristic, stratified by different graduation cohorts. The R^2 , or the percentage of overall variance explained by the predictor variables, is then used to determine the extent to which inequality on the basis of measurable institution-level characteristics, net of fields of study, is changing over time. This approach provides insight into whether institutional stratification is increasingly shaped by quantifiable dimensions of prestige or status.

Analyzing Changing Average Annual Earnings by Credential Over Time

The final component of horizontal stratification I examine is the credential itself. While earlier sections analyzed how each axis contributes independently to inequality, this section addresses a distinct but related question: are high-earning fields of study increasingly concentrated in high-earning institutions? This form of consolidation would suggest a deepening structural alignment between educational sorting and labor market inequality, and would echo recently described phenomena in the labor market wherein high-earnings occupations are more concentrated in high-earnings firms (Wilmer and Aepli 2021) and high-earnings workers are more concentrated in high-earnings firms (Song et al. 2019).

While the decomposition analysis in the first part of the analysis tests whether individuals are shifting across fields of study and institutions by recovering the level of overall variation due to the interaction effect of these two axes of horizontal stratification, that is a slightly different question from what this proposed analysis addresses. This analysis analyzes whether fields of study themselves are differentially distributed across schools over time. Moving from having the analytical unit be at the level of the individual to

¹¹This is actually the same result as the university-level fixed effects from a two-way fixed effects regression such as that described in equation 4. It is simply rewritten here in this manner for clarity.

having it be at the level of the specific educational credential tests a distinct concept of institutional consolidation as opposed to more basic allocative and distributional forces.

To test the consolidation of high-earnings fields of study in high-earnings institutions, the co-location of low-earnings fields in low-earnings institutions, and how this is changing over time, I again return to the basic earnings equation in which credential-specific average annual earnings are operationalized as the additive effects of field of study-specific effects, institution-specific effects, and an interaction effect (equation 4). Using the fixed effects estimated for each field of study $F_{f,y}$ and institution $U_{u,y}$, I measure the changing correlation and covariance of each over time, weighted by cell size. By assessing the changing covariance of institutional and field-specific earnings effects, this analysis reveals whether horizontal stratification at the credential level is becoming more concentrated, creating a novel variety of inequality at a structural level, which is net of the other forms of inequality described in this paper.

Results

Understanding Changing Average Annual Earnings by Field of Study

The analysis in Part I showed two major findings. First, field of study is the primary axis of horizontal stratification with regards to annual earnings after five years of graduation, and its impact appears to be growing for successive graduation cohorts. Second, the growing influence of fields of study reflects shifts in their average annual earnings rather than changes in the distribution of students across specific credentials, institutions, and fields of study. So the question remains: what is causing changing average annual earnings by field of study to lead to greater between-field inequality?

Figure 5 shows the observed changes in the average annual earnings, adjusted for inflation, of bachelor's degree-holders five years after graduation, by aggregated field of study. There are clearly large discrepancies in which fields of study have secured earnings gains, with "Computer & Information Sciences" and "Mathematics and Statistics"

capturing over \$20,000 and \$10,000 in earnings increases, respectively. Likewise, fields of study like “Liberal Arts,” “Homeland Security & Law,” and “Education” have seen more modest decreases in average annual earnings. In this figure, I also show predicted changes in field of study average annual earnings based on changing field of study-to-industry flows alone, as measured in the data. As the figure shows, these predictions largely line up with observed changes in average annual earnings, indicating changing industry flows and changing industry-level average annual earnings play a potentially large role in explaining observed changes in average annual earnings by field of study. Importantly, the two fields of study “Computer & Information Sciences” and “Mathematics and Statistics” with the largest observed growth in average annual earnings are well-explained by industry. Likewise, “Liberal Arts,” “English,” and “Protective Services,” with the largest decreases in average earnings are also well-explained by industry. Some larger fields of study such as “Health Professions,” “Education,” and “Psychology” do not fit as perfectly, though it is worth noting that the direction of the change in earnings is correct for all but one field of study, “Architecture.”

*** Figure 6 About Here ***

These changes reflect a mixture of changing industry composition within the entire labor market sector composed of students with bachelor’s degrees, shifting industry flows within fields of study, and changing average annual earnings by industry for college graduates. An extended Kitagawa-Oaxaca-Blinder decomposition, shown in Figure 6, parses these differences in the predicted earnings changes. The re-composition of flows from fields of study to industry due to global shifts in industry representation among college-educated workers has little bearing on overall changes in field of study average annual earnings, as gains in certain industries are often offset by losses in others. Nevertheless, some findings are striking. Globally, there is clearly a shift away from “Educational Services” and “Public Services” towards “Professional, Scientific, and Technical Services,”

“Information,” “Health Care and Social Assistance,” and “Management of Companies and Enterprises.” There is also consolidation of lucrative industries net of overall shifts in industry flows for specific fields of study. “Professional, Scientific, and Technical Services,” “Information,” and “Finance and Insurance” industries are increasingly industry destinations for bachelor’s degree-holders with degrees like “Computer Science/IT Support,” “Mathematics and Statistics,” and “Business.” These same industries are less represented as destinations for degree holders coming from “Engineering,” “Engineering Technologies,” and “Liberal Arts” fields of study, among others. Otherwise, put, not only are there global changes in flows from school to industry for all Bachelor’s degree-holders that reflect a shift to a service economy rooted in white-collar industry, high-tech, and management services, but there is also a consolidation of these lucrative industries among a select subset of fields of study. This does not follow the typical RBTC and STEM/non-STEM split, as fields of study like “Engineering,” “Health Professions,” and “Biology” have been largely left in the lurch.

Moving away from changes in field of study-to-industry flows, it is clear that baseline linkages between fields of study industries coupled with changing industry-level earnings have contributed most meaningfully to changing average annual earnings by field of study. Fields of study with preexisting greater flows to industries that saw large increases in earnings obviously benefited more. Thus, once again “Computer Science/IT Support,” “Mathematics and Statistics,” and “Business” saw increased average annual earnings due to simple baseline connections to these industries, notwithstanding the global and local shifts towards these industries that they also saw.

Of course, changes in average annual earnings by field of study are not fully explained by shifts in industry placement or industry-level average earnings; the decomposition leaves a relatively small amount of earnings change unaccounted for by these factors alone. These may be due heterogeneity within two-digit CIP codes within smaller, more specific fields of study. For instance, the two-digit CIP code encompassing “Health Pro-

fessions” contains a wide variety of programs, with widely differing post-college outcomes and levels of perceived prestige. Likewise, there are certainly some synergistic earnings effects between fields of study and industry that cannot be captured here, due to the lack of industry-by-field of study specific earnings. Nevertheless, the residuals, for the most part, are much smaller than the overall changes, indicating that an industry-level explanation is a decent analytical lens for describing such changes over time.

In sum, several industries saw large increases in average annual earnings for college graduates over the period in question. Fields of study, due to baseline school-to-work linkages to these industries also saw large gains in average annual earnings by field of study. This complements global shifts among bachelor’s degree-holders into more lucrative industries and field of study-specific shifts into these lucrative industries. Of course, as this is an observational analysis, it is difficult to say that this effect is causal. Indeed, there may be large shifts in sorting into universities and field of study, complemented by a preference for these higher-quality workers by certain industries, resulting in increased compensation owing to levels of skills. However, such speculation and analyses are far beyond the scope of this paper. Nevertheless, I undertake an additional set of analyses (described and reported in the Appendix) to test whether the observed changes in field of study-level average earnings are better explained by shifting demographic composition within majors or by evolving industry flows. These models show that demographic change accounts for only a modest share of the observed trends (28%), whereas industry-based measures consistently explain the bulk (86%) of the variation (supplementary appendix Table A1). I also report predicted field of study-specific average earnings based on the two sets of predictors as a modified version of Figure 5 in supplementary appendix Figure A5. These findings reinforce the conclusion that the rising salience of field of study reflects the importance of field of study-to-industry linkages and industry earnings structures, with demographic compositional shifts in who enters particular fields being secondary.

*** Figure 7 About Here ***

Understanding Changing Average Annual Earnings by Degree-Granting Institution

Though between-university stratification only explains a small amount of overall variation in annual earnings among bachelor's degree-holders, it remains the first line of stratification for many high school students as they choose where to attend university. Figure 7 shows institution-level average annual earnings, after adjusting for the distribution of fields of study within educational institutions, regressed against five different continuous characteristics of degree-granting institutions for graduating cohorts 2001–2003 and 2013–2015. Remarkably, school characteristics associated with student quality and instruction like average SAT scores and the number of graduate and undergraduate students show stronger and steeper relationships with earnings over time. The relationship with indicators that may be more closely linked to concepts of pure prestige as opposed to quality, like tuition fees for in-state students and the admission rate, do not show the same patterns. In fact, undergraduate admission rate shows a weakening relationship with school premiums over time, though shifts in application behavior driven by application inflation, test-optional policies, and strategic enrollment management have potentially made admission rates a far noisier signal of institutional quality, weakening their relationship to earnings in ways not mirrored by other indicators. While all of these characteristics reflect both latent concepts of quality and prestige to some extent, these patterns are suggestive and merit further exploration in future research.

The strongest and most dynamic relationship is that of average SAT Score and institution-level average annual earnings, with the R^2 increasing from 0.12 to 0.35 between the two cohort spans. Likewise, the variance explained by the logged number of graduate students increased from 0.05 to 0.12 and the logged number of undergraduates increased from 0.01 to 0.05. Of course, these relationships are associational only, and

there is no causal component to this part of the analysis. However, it could easily be proposed that degree-granting institutions are becoming more stratified along observable characteristics. While between-school disparities in outcomes is not a main driver of overall inequality in the analyses in Part I, it is nevertheless striking that the institutional differences we can observe are becoming more tightly aligned with dimensions that sociologists have long associated with stratification. Further, this relationship is not mechanically caused by the inflation of university characteristics over time and changing leverage of certain data points. Repeating the analysis holding constant university characteristics by taking their mean value across all cohorts and simply allowing university premiums to change shows substantively similar results (Appendix Figure [A3](#)).

Finally, this evidence of the changing interrelationship of the characteristics of educational institutions and institutions' average outcomes calls into the standard practice of grouping educational institutions based on observable characteristics. Although the relationship between observable characteristics and institutional earnings has grown stronger over time, these features still explain only a modest share of the variation. Most of the inequality in institutional outcomes remains unexplained by commonly used metrics, suggesting that the deeper sources of institutional stratification lie in less visible organizational processes, historical positioning, and localized industry linkages. This final point should also give researchers pause when using survey data to assess horizontal inequality among institutions, as it is clear that outcomes are highly variable even among nominally similar institutions. As stated in Part I, this finding also reinforces that the PSEO dataset is itself quite informative in describing between-institution inequality, for even though it is not representative by itself of the full American ecosystem of higher educational institutions, the variation between institutions does not follow clear or stable patterns, which suggests that placing too much emphasis on exact representativeness is not especially productive given the absence of a full census of institutions.

*** Table 6 About Here ***

Understanding Changing Average Annual Earnings by Credential

The final research question moves beyond the analysis of field of study effects and institution effects in isolation to look at their intersection. Specifically, this analysis answers the question of whether high- or low-earning fields of study are increasingly consolidated in high- or low-earning institutions. Table 6 shows that this is not the case. To begin with, both the correlation and covariance of institutional effects and field of study effects are close to zero, indicating there is little baseline consolidation. Furthermore, there is no discernible pattern over time in either of these indicators. Thus, while specific credentials from specific academic departments explain 3–4 percent of overall earnings variation, there is little evidence that the dynamics of how it comes about are being dramatically reorganized.

Discussion

This study provides new evidence on how horizontal stratification among bachelor's degree recipients has evolved over time. Strikingly, variation in pay is becoming more explainable by degree characteristics. Of the three primary axes examined—field of study, institution, and their intersection—field of study stands out as the most important for earnings inequality. From graduating cohorts from 2001 to 2015, it explains a growing share of post-graduation earnings variation, while institutional and credential effects remain smaller and more stable.

The growing importance of field of study as a source of horizontal stratification is not explained by the reallocation of students across majors or institutions, nor by changes in the number or distribution of credentials and fields of study alone. It also appears not to be primarily a function of changing demographics within fields. This does not invalidate ample research showing that fields of study are gendered and racialized, impacting their value on the labor market, especially in the patterned ways that they lead to gendered

or racialized occupations (e.g., England et al. 2007; Alon and DiPrete 2015; Goyette and Mullen 2006; Dalberg et al. 2024). Rather, it indicates that the *changing* importance of degree characteristics described in this paper are not driven by *changes* in these forces. Substantively, the results suggest that broader structural shifts in the labor market, particularly the expansion of high-earnings service industries like technology, finance, and professional services, may be contributing to the rising returns associated with certain fields of study. Notably, this transformation does not uniformly reward traditionally “technical” majors like engineering or health sciences. Fields such as business, social sciences, and communications/journalism, which are often peripheral in frameworks based on RBTC, have experienced substantial earnings growth, largely due to their alignment with high-paying sectors whose earnings structures continue to diverge from the rest of the labor market. These patterns imply that credentials increasingly derive value from where they are positioned in an evolving industrial earnings structure, not simply from changes in student composition. Although the analyses focus on explaining changes over time in the contribution of institutions and fields of study to earnings inequality, it is important to note that point-in-time differences across credentials may still reflect demographic, compositional, or selective forces operating within a given cohort. The key result here is that these forces do not appear to be the drivers of the secular rise in field of study-linked inequality, even if they may continue to shape static gaps at any moment in time.

These trends are visible due to the recency and granularity of the data, which capture labor market outcomes through 2020. This extended temporal scope is critical, as it allows the analysis to include the post-recession period, a time marked by deep restructuring in the labor market, the maturation of the tech sector, and growing divergence in industrial earnings structures. Unlike earlier periods shaped by the initial waves of changes due to computers, the post-2008 landscape reflects a new stage of labor market change, in which a narrower set of industries increasingly concentrates economic

returns. In most cases, observed earnings gains reflect rising earnings within industries already associated with particular fields of study. In others, they stem from a tighter clustering of lucrative sectors around a smaller subset of fields of study. This dynamic suggests that education policy, career advising, and labor market interventions must grapple with a world in which the labor market value of a degree is increasingly determined not by content alone, but by how credentials are absorbed into an evolving and uneven industrial structure.

Universities themselves account for only a small amount of variation in annual earnings, and yet this variation is increasingly explainable based on observable university characteristics. This finding contrasts with the findings of Borgen and Mastekaasa (2018), who finds that universities play no role in dictating labor market outcomes in excess of the individual department (specific educational credential). This discrepancy is likely linked to comparative differences in the U.S. and Norwegian context and motivates future work on school-to-work linkages in a comparative setting. Though their analyses include individual-level characteristics, they report supplementary analyses that omit such characteristics that are consistent with the key assumptions of this research based on the synthesis of Bleemer and Quincy (2025): differential sorting into programs does not appear to attenuate effects of fields of study, and it only slightly attenuates the effects of institutions.

The two-pronged increase in stratification along the axes of fields of study and institutions makes salient an increasingly important source of inequality in society—what and where one studies during their bachelor’s degree—but this increasing stratification in outcomes opens doors for increased inequality to come about due to other, allocative factors. For instance, differential sorting into different fields of study on gendered and racialized lines (e.g., Lepage, Li, and Zafar 2025) would exacerbate known sources of horizontal stratification based on selection into fields of study and institutions. Further, research has documented a pattern of an increased prevalence of GPA-restricted

majors within colleges, and these internal gatekeeping mechanisms limit access to high-return fields, often along lines of prior preparation and social background, reinforcing inequality even within institutions (Bleemer and Mehta 2024). Increasingly divergent returns across fields of study only heighten the stakes of such mechanisms, as access to lucrative majors then becomes an additional channel through which educational systems reproduce broader social inequalities.

Although specific credentials account for a similar share of overall earnings inequality as educational institutions, there is no clear evidence that high- or low-earning fields of study are becoming increasingly concentrated within correspondingly high- or low-earning institutions, respectively. The overall distribution of majors across institutions has remained remarkably stable with some exceptions, suggesting that the growing influence of field of study is not being driven by credential-level consolidation. Nor do patterns of field emergence and obsolescence appear confined to particular types of institutions. Instead, the rise in horizontal stratification reflects broader changes in the economic value of fields themselves, shaped by evolving industrial earnings structures and shifting pathways from education to work. These findings not only clarify the empirical contours of horizontal stratification but also raise important questions for how sociologists conceptualize the education–labor market relationship moving forward.

Finally, an enduring concern is selection: to what extent do the observed differences in fields of study, institutions, and credentials reflect value added versus sorting by ability, preferences, or background? While the design here does not identify causal effects, three pieces of evidence help calibrate interpretation of my findings. First, the demographic fixed-effects analyses in the supplementary appendix indicate that within-field shifts in gender and race/ethnicity explain only a modest share of changes in field premiums, whereas industry-linked measures account for substantially more variation over time. Second, robustness checks that reweight institutions and reorder fixed effects suggest that the growing role of fields is not just an artifact of compositional shifts in the mix

of fields of study across campuses. Third, recent causal evidence implies that field of study differences largely represent value added (on the order of 100%), while institutional differences are attenuated by selection but still predominantly causal (roughly 80%), with estimates for institutions consistent with Chetty et al. (2023) and Bleemer and Quincy (2025). Finally, recent qualitative work by Moss-Pech (2025) and Streib (2023) points in a similar direction, albeit in distinct settings. Moss-Pech’s study of majors at a large Midwestern flagship university shows that within-institution stratification by field of study does not primarily reflect differences in ambition or talent. Rather, between-field gaps in employment outcomes are mediated by how majors connect students to particular sectors and kinds of entry-level jobs. Streib’s analysis of business graduates from a non-elite public university likewise finds that early pay is only weakly tied to students’ class-based resources and more to opaque, luck-driven hiring processes. Taken together, these studies support the premise that observed differences across fields are shaped less by underlying student characteristics than by how credentials are positioned vis-à-vis employers and sectors of work. Thus, while the evidence in support of the key assumptions employed in my analyses is robust to multiple specifications across several papers, more evidence using large scale administrative data like what is used in this paper would be welcome, especially since the importance of selection may change over time. Nevertheless for the time being, the decomposition results should be read as nearly equal to the causal importance of fields of study and as an upper bound for institutions. Substantively, selection concerns do not overturn the main conclusion: the dominant driver of horizontal inequality lies in field-linked positioning within an evolving industrial earnings structure; institutional effects remain secondary, and their interpretation should be tempered only slightly by concerns over selection processes.

Implications for Research on Education and Labor Market Inequality

These findings suggest that sociological research on education and labor markets must more fully integrate perspectives from the school-to-work literature and from research on labor market polarization, firm segregation, and industrial change (Wilmer and Aeppli 2021; Godechot et al. 2024; Haltiwanger et al. 2024). The increasing differentiation of earnings across fields of study aligns with evidence that earnings, capital, and organizational power are consolidating within a narrow set of high-value industries. Rather than treating education and industry as separate domains, future work should consider how stratification across credentials and stratification across industries are increasingly intertwined, with fields of study functioning as channels into specific segments of a restructured labor market.

Evidence that industry dynamics are central to the evolution of horizontal stratification also highlights the need to examine the institutional mechanisms that reinforce or mitigate these patterns. Universities, firms, and occupations all participate in constructing and maintaining the links between particular credentials and particular sectors. Relational inequality theory (Tomaskovic-Devey and Avent-Holt 2019) provides a starting toolkit for considering how organizational ecologies shape these linkages, by directing attention to how resources, authority, and rewards are distributed through ongoing relations between educational units and employers rather than through isolated attributes of either side. Applying this perspective to field-to-industry connections would help clarify when credentials simply track preexisting hierarchies and when they actively participate in producing new ones.

The results also speak to theories of educational stratification that emphasize qualitative differentiation within expanding systems. They are broadly consistent with the Effectively Maintained Inequality framework (Lucas 2001), which posits that advantaged groups secure higher-quality credentials as access expands, and the growing importance

of field of study documented here suggests that horizontal distinctions have become a key arena in which such qualitative advantages are accumulated. Similarly, the findings can be seen as a contemporary elaboration of Collins’s theory of credentialism (Collins et al. 1979). Rather than operating only at the level of broad degrees or licenses, closure processes now appear at a finer grain, as specific institution–field combinations regulate access to distinct parts of the labor market. Horizontally stratified credentials thus form an infrastructure through which symbolic and skill-based distinctions are converted into unequal economic outcomes.

Finally, the growing importance of horizontal stratification raises pressing policy concerns. As earnings gaps across fields and institutions widen, expanding access to college in general is unlikely to reduce inequality unless students from marginalized backgrounds also gain access to higher-return majors and more advantaged institutions (Bleemer and Mehta 2022; Monaghan and Jang 2017). Internal gatekeeping mechanisms such as GPA-restricted majors and selective program admissions may become increasingly consequential as field-based returns diverge, especially if they track prior preparation and social background. At the same time, the strengthening association between institutional characteristics, such as selectivity or graduate enrollment, and graduate earnings suggests that institutional prestige may play a growing role in organizing opportunity. These trends underscore the need to evaluate policies not only in terms of overall college access, but also in terms of who is able to enter, persist in, and benefit from the specific credentials that carry the greatest labor market returns.

Limitations

To contextualize my findings and guide future research, it is important to acknowledge several limitations, some of which have been alluded to in earlier sections. First, the analyses rely on summary statistics rather than individual-level data. Although these statistics are highly granular, capturing credential-cohort combinations with as few as

30 graduates, they necessarily obscure some within-cell variation and do not allow for the identification of micro-level selection mechanisms or causal pathways related to individual characteristics such as race, gender, parental education, or academic preparation. Nonetheless, sensitivity checks (Appendix Figure A4) suggest that the patterns identified are robust across specifications that may be related to such mechanisms. Moreover, this design tradeoff is what enables the study to provide a rare, macro-level perspective on how horizontal stratification unfolds across institutions and fields of study over time.

Second, this analysis focuses on three analytically tractable forms of horizontal stratification: institution, field of study, and their intersection. Other important dimensions such as intra-field specialization, co-curricular experiences, and informal academic tracking are not examined here, though they likely shape labor market outcomes and interact with the patterns observed. Rather than cataloging all forms of horizontal inequality, this paper traces how credentials have become increasingly differentiated in economic value amid structural labor market change. Future research should explore these finer-grained dynamics as richer data become available.

Third, as noted in the results, Ivy League and similarly highly-selective private institutions are underrepresented in the dataset. While the broader underrepresentation of private universities is addressed through compositional reweighting and sensitivity checks, the absence of Ivy League and their peer institutions poses a distinct limitation. Nevertheless, these schools enroll relatively few students compared to large public universities, so their exclusion likely has minimal impact on overall trends. However, it does constrain generalizability at the top of the institutional earnings distribution, where the labor market returns to elite prestige may be most pronounced. As a result, this study may understate the extent of institutional stratification at the highest levels of the college hierarchy, but it is likely broadly representative of the remaining 98% of higher educational institutions existing outside of the elite strata.

Focusing on annual earnings five years after graduation also has limitations. While

robustness checks (Appendix Figure A2) show similar static patterns of horizontal stratification at one and ten years and comparable dynamic trends at one year, income is an imperfect proxy for social position, well-being, and long-term mobility. Still, it remains a uniquely valuable measure: it reflects actual economic standing, avoids measurement bias, and is especially relevant for recent graduates navigating economic precarity. For many, particularly those from non-elite backgrounds, earnings serve as a proxy for broader labor market outcomes. Early-career income also aligns with experiences common to this key life stage, such as student loan repayment and household formation, making it a salient indicator of inequality. Other dimensions of stratification like non-monetary compensation, occupational prestige, or long-term growth are harder to measure (Cheng 2014; Cheng and Song 2019) and likely correlate with earnings, suggesting my results may understate overall horizontal stratification. The data also exclude individuals still in school. Those who later complete graduate degrees enter the analysis only upon labor market participation, making it difficult to isolate the effects mediated through the pursuit of an advanced degree. Nevertheless, sensitivity checks affirm that the main findings are robust to different periods of follow-up post-graduation. Future research with richer labor market indicators and linked microdata would help clarify these mechanisms.

Additionally, this analysis cannot disentangle how much of the observed stratification reflects worker–firm sorting. Because firms are nested within industries, rising industry premiums could partly reflect shifts toward higher-paying firms within those industries rather than changes common to all firms. And since the PSEO data used in these analyses do not include firm identifiers or characteristics, one cannot analyze firm-level phenomena such as wage-setting or recruiting pipelines. Access to microdata linking education records to employer-level data would clarify the extent to which horizontal stratification reflects institutional value added versus sorting into higher-earnings firms. Using AKM models to disentangle worker-level skill sets, firm-level earnings premiums,

and employer-employee match effects is a natural next step when the data become available (Abowd, Kramarz, and Margolis 1999).

In addition, this analysis necessarily excludes individuals who begin but do not complete a bachelor's degree. As noted earlier, the study defines horizontal stratification as variation within the population of bachelor's degree recipients, which allows for clear analytical separation from vertical processes such as degree attainment. As a result, the findings reflect patterns of inequality among graduates only, and do not capture the potentially large disparities generated by differential rates of completion by field of study and institution. Because non-completion is strongly patterned by background characteristics and institutional context, the earnings differences reported here are likely conservative. Future research should examine how institutional and field-level differences in retention and completion interact with vertical stratification and shape access to the types of credentials studied here.

Finally, there is question of industry as the appropriate labor market stratifying force to explain the inequalities described here. Other indicators, like RBTC at the occupation level or deindustrialization more broadly are alternative lenses through which this analysis could have been performed. While the data limit me to focusing on industrial change, I believe this is also a principled choice. Other research has pointed out the realignment of industries in the labor market as a primary source of changing dynamics of inequality (Wilmers and Aepli 2021; Haltiwanger et al. 2024). Wilmers and Aepli (2021) in particular outline how changing occupational dynamics coincide with changing industry dynamics, though the question of which precedes the other remains an open question that should be addressed with further research.

Future Research Directions

While this study provides a broad overview of the changing landscape of horizontal stratification in higher education, it also raises several key questions that future research

should address. One of the most pressing concerns is the causal mechanism behind the school-to-work linkages discussed here, and their knock-on effects for selection into college and fields of study in the first place. This study demonstrates that industry-level dynamics play a crucial role in shaping earnings stratification across fields of study, but it does not pinpoint whether these effects stem from employer preferences, student self-selection, or university-side institutional steering.

Future work should examine how these trends vary across demographic groups. Prior research suggests that access to high-return majors like STEM and finance is stratified by race, gender, and class (Xie and Shauman 2003; Gaddis 2015). As the field described here increasingly shape earnings, horizontal stratification may reinforce existing inequalities. Investigating whether industry practices or educational interventions can offset this stratification is essential.

An additional avenue for future research involves examining the role of school-to-work linkages themselves. Prior work has shown that stronger linkage systems, typically analyzed in cross-national comparisons, shape employment outcomes by more tightly connecting educational credentials to specific labor market destinations (DiPrete et al. 2017; Bol et al. 2019). While this study does not center on linkage strength, the fine-grained credential-level structure of the data, combined with detailed information on industry destinations, offers a novel opportunity to do so. This creates the possibility of extending the original framework to examine variation in linkage strength across institutions, fields of study, and credentials within the U.S. context. Such an approach would allow researchers to test whether more tightly coupled credentials produce more stable or more unequal outcomes in a causal framework, and to investigate how linkage strength itself evolves alongside changes in industrial structure and institutional positioning.

Graduate education also warrants closer attention. While this study focuses on bachelor’s degree holders, similar patterns in horizontal stratification may be simultaneously occurring, or even intensifying, at the master’s and doctoral levels. As advanced degrees

are often viewed as mobility pathways, understanding whether certain graduate credentials are consolidating economic advantages while others lag is critical. In addition, future research should explore how post-baccalaureate education moderates the patterns observed here, particularly for those whose earnings outcomes reflect the combined value of undergraduate and graduate training.

Finally, a comparative perspective is necessary to assess to what extent these trends are unique to the United States. The U.S. higher education system is characterized by its flexibility and relatively weak school-to-work linkages compared to other nations, where vocational training and credentialing structures may create different trajectories for students. If similar patterns of growing horizontal stratification are observed in other countries, it may suggest a global strengthening of school-to-work linkages due to specific industrial relations rather than being due to the specific institutional features of U.S. higher education. Indeed the only study remotely comparable to this, which was conducted in a Norwegian context, found that there was no effect of educational institutions in excess of credential-specific effects, highlighting the importance of national context (Borgen and Mastekaasa 2018). Future research should explore cross-national comparisons to determine whether the increasing importance of field of study and institutional prestige in shaping earnings outcomes is a universal phenomenon or whether it is shaped by national policies, labor market structures, and educational institutions.

Conclusion

This paper set out to accomplish the seemingly simple task described in its title. It asked whether what students study or where they study matters more for structuring earnings inequality among bachelor's degree recipients. I argue that it has succeeded in doing so. I showed that fields of study account for a large and growing share of the variation in earnings among graduates, that institutions account for a smaller but still meaningful share, and that the specific combinations of fields and institutions add rela-

tively modest additional structure. I also showed that these patterns reflect evolving ties between fields of study and industries rather than shifts in enrollments or demographic recomposition alone. Yet in answering the initial question, the analysis also brought into view a broader set of processes that extend beyond a simple comparison of what and where. The findings here suggest that horizontal stratification is not a fixed hierarchy of majors or colleges but a moving configuration that responds to changes in labor markets, institutional choices of course offerings, linkages between schools and work, and the self-reinforcing dynamics that exist among them. This evidence provides a foundation for future work that traces how these shifting credential hierarchies intersect with demographic inequalities, politicized incentive structures in both labor market and higher education policy, and organizational dynamics in hiring.

Although the increasing horizontal stratification described in this paper raises concerns about inequality, it could plausibly also be interpreted as reflecting a labor market that rewards acquired skills and productive specialization more directly than one centered on institutional prestige or individual characteristics. In this sense, the rising importance of what graduates study relative to where they study may indicate that economic outcomes increasingly track substantive competencies and human capital rather than reputational advantages. If this interpretation holds, the growth of field of study-based stratification could represent a partial shift toward a more meritocratic linkage between education and work, where skill-based differentiation outweighs credential-based gatekeeping. Further study of the mechanisms conferring field of study-level differences in earnings is needed, however.

A central contribution of this study lies in its use of temporally rich, credential-level data that span over fifteen years and include a variety of U.S. higher educational institutions. Prior research has often been limited to single cohorts, broad institutional groupings, or narrow timeframes. By contrast, this study captures how stratification unfolds dynamically, in tandem with structural shifts in the labor market. This level of

temporal and institutional granularity allows for a more empirically grounded account of how educational inequality is produced and reproduce, not just through individual sorting, but through evolving relationships between education and work.

These findings point to the need for a reconceptualization of how educational stratification interacts with broader processes of labor market change. Horizontal and vertical stratification cannot be understood merely as reflections of individual preferences or institutional sorting mechanisms, but as parts of a dynamic system in which educational and economic structures co-evolve. As the boundaries between fields of study and labor market positions become more tightly organized by industry-level transformations and other labor market processes, new pressures are placed on higher education systems that were built for an earlier era in which broad human capital accumulation was presumed to guarantee stable employment. This is increasingly at odds with a labor market reshaped by technological advances, shifting expectations about skills, and growing uncertainty about what comes after the college-for-all era. Understanding the co-evolution of specific credentials, industry demands, and emerging skill regimes will be essential for advancing theories of education, stratification, and inequality.

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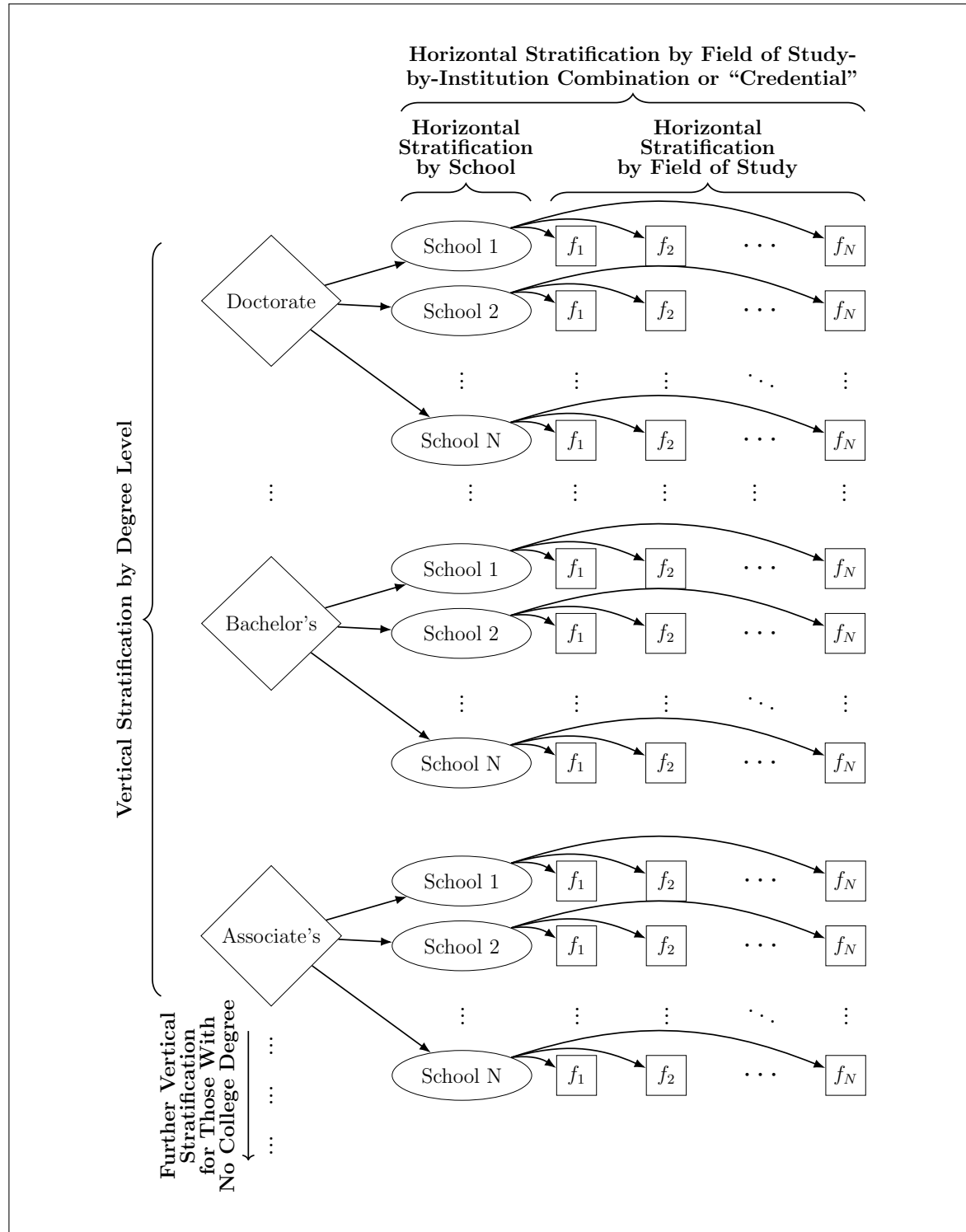


Figure 1: Schematic Showing Vertical Stratification and Horizontal Stratification by Both School and Field of Study

Notes: Only vertical stratification at the two-year college level and above is shown. There is unshown vertical stratification for those with high school, less than high school, and no schooling. Whether or not these exhibit meaningful horizontal stratification by school is unknown, and there is no field of study specialization below the college level.

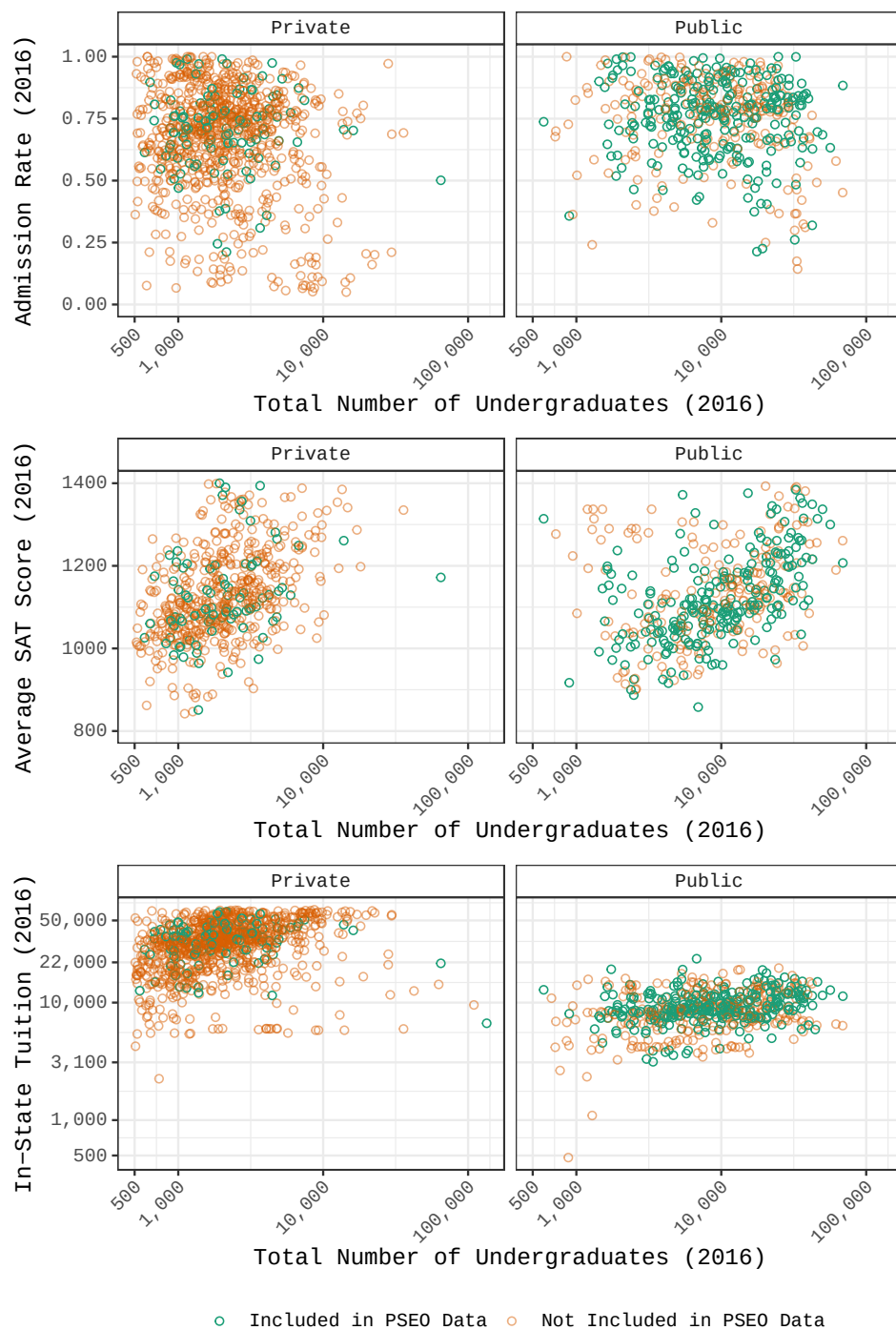


Figure 2: Scatter Plot of All Universities and Universities Included in PSEO Data, By Total Number of Undergraduate Students, In-State Tuition, Admission Rates, and Average SAT Scores

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes; The U.S. Department of Education's College Scorecard.

Notes: Only universities are shown that have at least 500 undergraduate students.

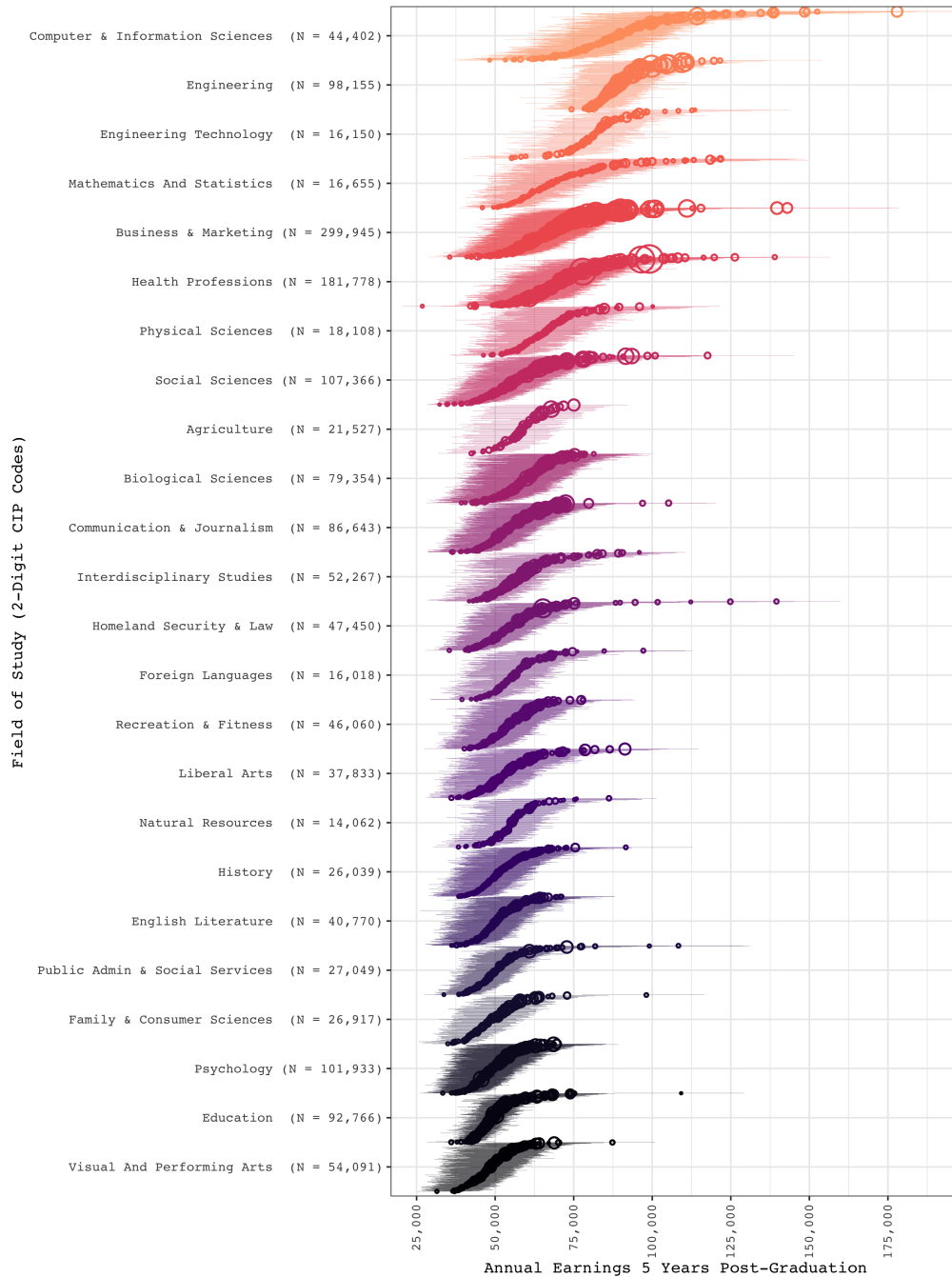


Figure 3: Distributions of Average Annual Earnings, Five Years Post-Graduation, for Specific Credentials (Fields of Study Within Specific Degree-Granting Institutions), By Aggregate Field of Study; Students Graduating 2013–2015

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes.

Notes: Groupings refer to CIP codes aggregated at the two-digit level, though each data point represents a credential (degree-granting institution and field of study combination) at the four-digit level. Colors are to help visually distinguish fields of study.

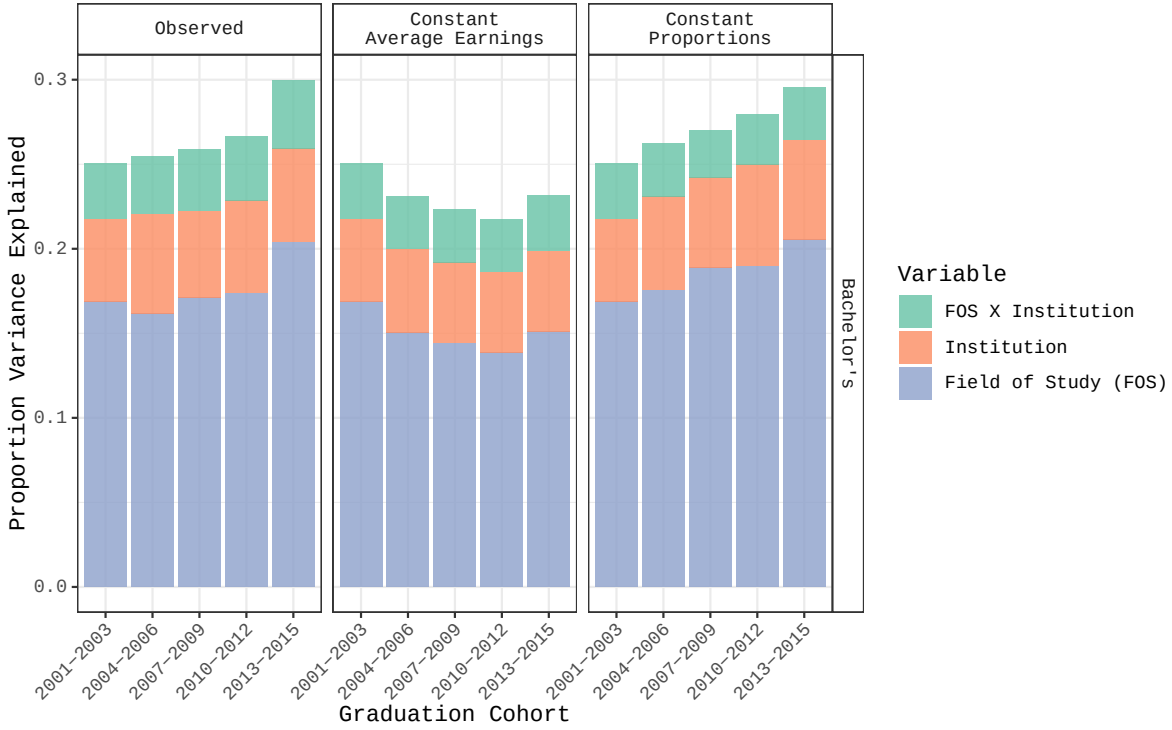


Figure 4: Variance Decomposition of annual earnings Five Years Post-Graduation by Field of Study (Major), Educational Institution, and Their Intersection Alongside Scenarios Assuming Constant Allocations Across Fields and Universities or Constant Average Annual Earnings Across Fields and Universities

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes.
Notes: “Constant Proportions” refers to the scenario in which all enrollments by field of study and degree-granting institutions are held constant at their 2001–2003 numbers, and “Constant Average Earnings” refers to the scenario in which enrollments are allowed to vary, but credential-specific average earnings are held constant. Results reflect data reweighted to more closely represent all U.S. universities and colleges.

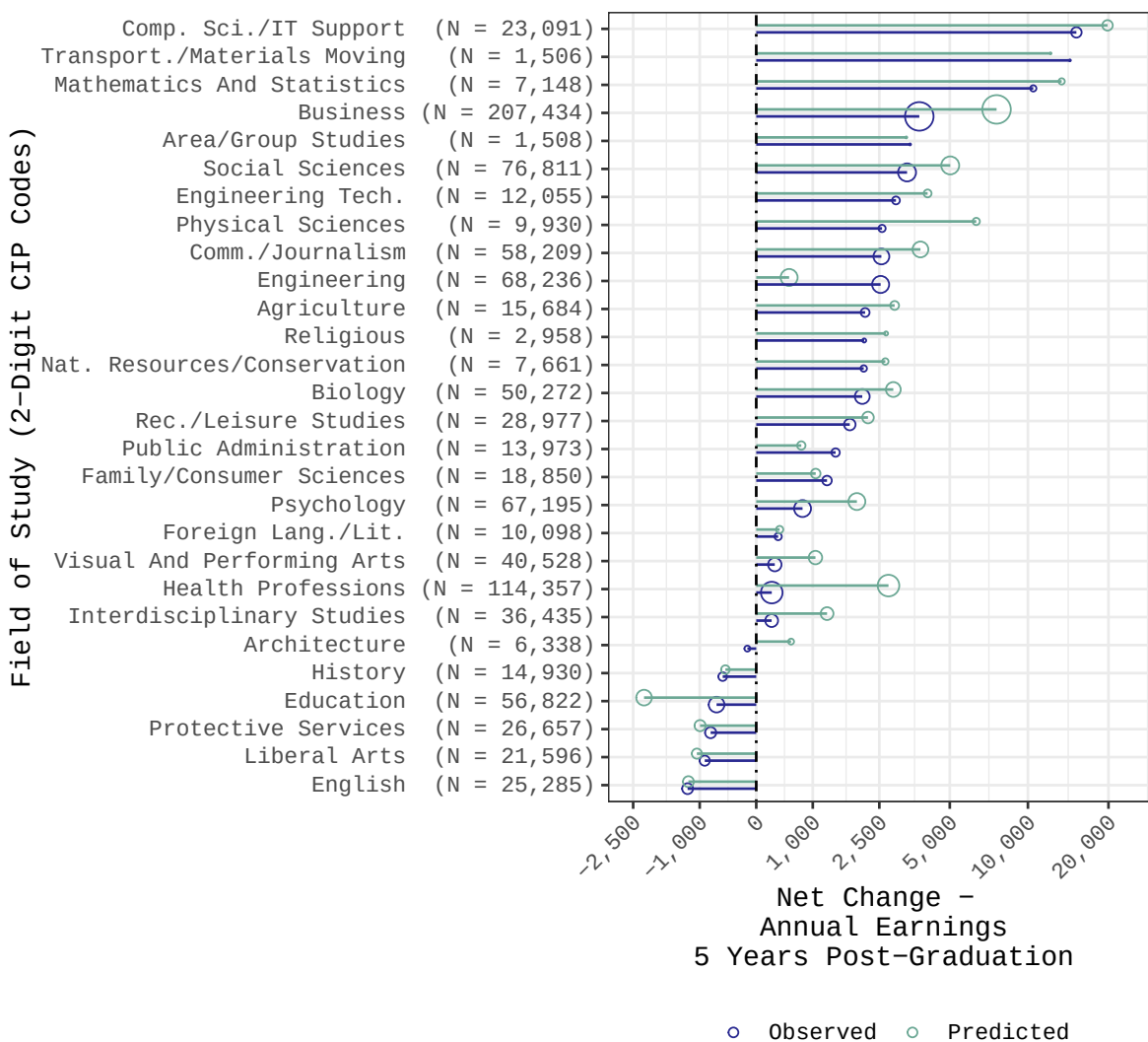


Figure 5: Change In Observed Average Field of Study (Two-Digit CIP Codes) Annual Earnings Five Years Post Graduation, Comparing Cohorts 2013–2015 to Cohorts 2001–2003

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes; The American Community Survey.
Notes: Only fields of study with at least 1,000 graduates per year (3,000 per cohort span) are shown. Size of circles is correlated with the number of students graduating in each cohort. ‘N =’ for each cohort refers to their absolute sizes for graduating cohorts 2013–2015.

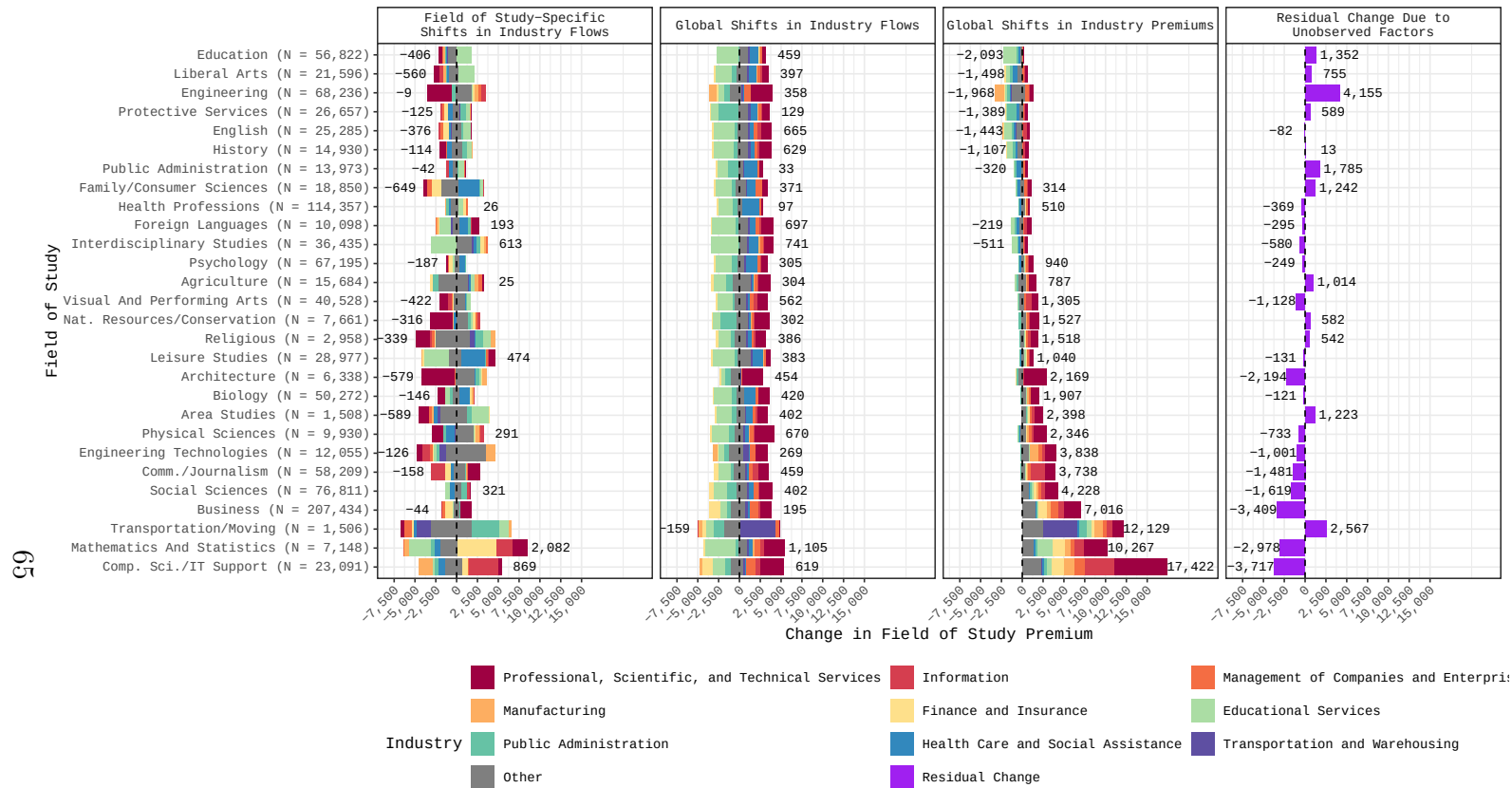


Figure 6: Kitagawa-Oaxaca-Blinder Decomposition of Observed Changes Between 2001–2003 and 2011–2013 Graduates in Field of Study average annual earnings at the Bachelor's Level, Using Observed Changes in FOS-Industry Flows Changes in Calculated Industry-Level Average Annual Earnings

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes, The American Community Survey.

Notes: Numbers next to each set of stacked bars represent the total change in average earnings for a given field of study, summed across all industry contributions of a given kind, be they positive or negative. The nine industries with the largest contributions are shown explicitly, while the others are labeled "Other."

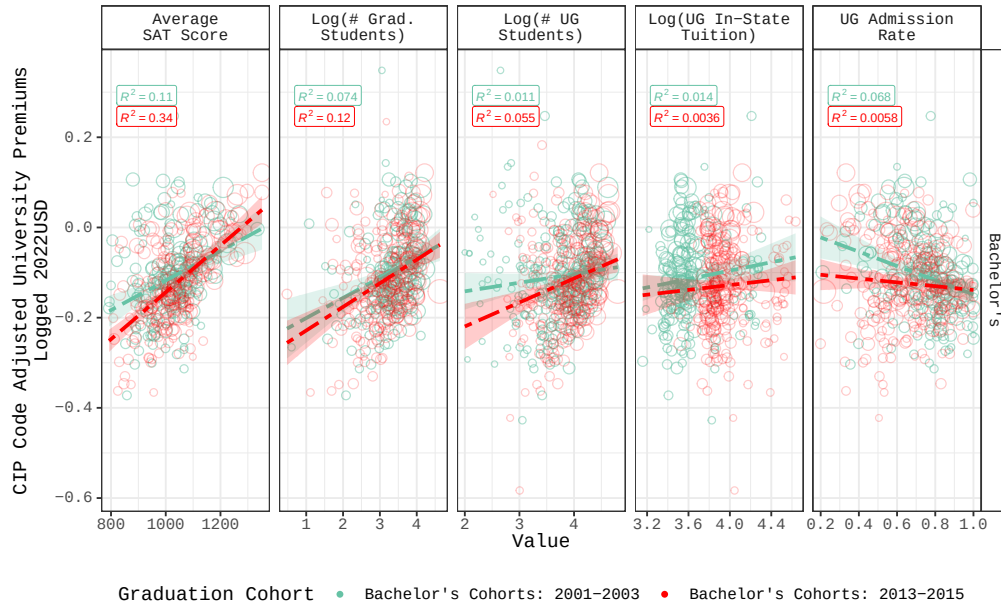


Figure 7: Degree-Granting Institution Average Premiums, After Adjusting for Distribution of Fields of Studies, Regressed on Institutional Characteristics for Two Cohorts

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes; The U.S. Department of Education's College Scorecard.

Table 1: Table displaying exemplars of existing literature on horizontal and vertical stratification in income based on education, separated into point-in-time and change-over-time accounts.

		Within Cohorts (point-in-time)	Across Cohorts (change-over-time)
Vertical Stratification (between degree-level)		• Heterogeneous returns to college (Brand and Xie 2010) • College as equalizer (Torche 2011; Zhou 2019)	• Education–technology race framework (Goldin and Katz 2008) • Task polarization and wage trends (Autor 2014) • College premium plateau (Autor et al. 2020)
Horizontal Stratification (within degree-level)	Field of study	• Causal field of study effects, Norway (Kirkeboen et al. 2016) • field of study channeling and stratification (van de Werfhorst 2002) • Earnings by major (Carnevale et al. 2013) • <i>*See bottom-right cell.*</i>	• Changes in College Skills and the Rise in the College Wage Premium (Grogger and Eide 1995) • <i>*See bottom-right cell.*</i>
	Institution	• Effects of elite college attendance (Brewer et al. 1999) • Estimating the returns to elite colleges (Dale and Krueger 2002) • Effects of college prestige (Brand 2006) • <i>*See bottom-right cell.*</i>	• Changes in college match in NLSY (Dillon and Smith 2020) • <i>*See bottom-right cell.*</i>
	Field & institution	• Horizontal Stratification in Norway (Borgen and Mastekaasa 2018) • Institution-, Major-Specific Premiums in Ohio (Ost et al. 2024) • <i>*See bottom-right cell.*</i>	• Changing Regressivity of College Degree (Bleemer and Quincy 2025) • Changing Horizontal Stratification (<i>The present study</i>)

Notes: * Change-over-times accounts necessarily always overlap with point-in-time accounts. Likewise, accounts of horizontal stratification by field of study and institution always overlap with accounts separately by field of study or institution. Studies that do so are only included in their most specific cell in the above table to avoid needless repetition.

Table 2: Characteristics of PSEO Sample, All U.S. 4-Year Colleges, and the Reweighted PSEO Sample, By Institution

	Admission Rate	Average SAT	In-State Tuition	Graduates	Undergraduates	% Private
Target	71.3 %	1,169	16,612	1,231	7,643	38.9 %
Sample	74.3 %	1,127	12,939	3,276	11,894	14.7 %
Reweighted	70.6 %	1,161	13,197	1,279	8,176	34.0 %

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes.

Notes: Methods for weighting data are detailed in the supplementary appendix.

Table 3: Summary Characteristics of Workers Five Years After Graduation Included In Data

Graduation Cohort	Individuals	Schools	FOS (4 Digit)	FOS (2 Digit)	FOS-School (4 Digit)	FOS-School (2 Digit)	Average Salary (2022USD)
Full Data							
2001-2003	586,872	275	227	34	4,890	2,849	58,427
2004-2006	902,432	338	251	36	6,310	3,620	57,494
2007-2009	932,181	376	256	36	7,379	4,156	55,410
2010-2012	1,134,101	434	273	36	8,888	4,941	58,497
2013-2015	1,309,057	460	274	36	9,808	5,387	61,311
Restricted Set							
2001-2003	579,723	270	227	34	4,842	2,818	58,354
2004-2006	801,515	270	247	36	5,523	3,134	57,342
2007-2009	746,814	270	247	35	5,703	3,197	55,310
2010-2012	818,058	270	258	35	6,161	3,378	57,967
2013-2015	900,124	270	261	36	6,635	3,555	60,646

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes.

Notes: The restricted data set only contains those schools which are present in all five cohorts.

Table 4: Top and Bottom Ten University-Fields of Study Combinations by Median Earnings Five Years Post-Graduation

University	Field of Study	Earnings Rank	Median Annual Pay (2022USD)	25th and 75th Annual Pay Percentiles (2022USD)	Number Graduates	% of Grads in Major Ind.
Top and Bottom 10 Credentials - Bachelor's - Graduates 2001-2003						
TX Tech Univ.	Engineering	1	192,914	[150,091, 250,007]	32	0.32
TX A&M Univ.	Engineering	2	166,293	[128,850, 220,166]	99	0.30
Univ. of TX - Austin	Engineering	3	164,801	[126,484, 220,373]	66	0.32
Colorado School of Mines	Engineering	4	156,629	[125,880, 213,300]	53	0.40
Montana Technological Univ.	Engineering	5	153,211	[108,199, 197,490]	52	0.27
OR State Univ.	Pharmacy	6	150,091	[135,781, 166,639]	54	0.33
Purdue Univ.	Pharmacy	7	142,605	[117,974, 163,485]	114	0.53
LA State Univ. & A&M Colg.	Engineering	8	140,944	[113,341, 171,849]	35	0.31
Univ. of LA at Monroe	Pharmacy	9	137,486	[119,561, 160,715]	33	0.85
Univ. of Montana (The)	Pharmacy	10	132,068	[112,895, 152,543]	145	0.48
Fort Lewis Colg.	Anthropology	4,881	32,304	[24,601, 43,164]	31	0.16
North Greenville Univ.	Religious	4,882	32,304	[25,573, 41,247]	58	0.19
Kent State Univ.	Drama/Theatre	4,883	32,219	[25,944, 44,441]	49	0.16
Univ. of OR	Ethnic, Cultural Minority, Gender, And Group Studies	4,884	32,004	[25,476, 45,923]	53	0.29
McNeese State Univ.	Fine And Studio Arts	4,885	31,462	[25,726, 46,077]	30	0.31
LA State Univ. & A&M Colg.	Drama/Theatre	4,886	30,499	[22,705, 47,315]	38	0.18
Univ. of Northern Colorado	Music	4,887	30,132	[21,188, 49,171]	59	0.25
Shenandoah Univ.	Drama/Theatre	4,888	29,535	[18,267, 43,942]	63	0.31
Univ. of WI - Madison	Music	4,889	29,471	[19,726, 52,390]	42	0.18
Univ. of Colorado Boulder	Religious	4,890	27,929	[20,809, 41,247]	31	0.23
Top and Bottom 10 Credentials - Bachelor's - Graduates 2013-2015						
Univ. of IL Urbana-Champaign	Computer Science	1	166,721	[112,653, 254,204]	602	0.40
Univ. of IL Urbana-Champaign	Math and Comp. Science	2	159,898	[115,503, 215,797]	48	0.27
CUNY York Colg.	Health Professions	3	155,421	[123,532, 180,099]	66	0.63
Univ. of MI	Engineering	4	152,543	[105,327, 217,603]	185	0.32
TX A&M Univ.	Engineering	5	152,452	[100,629, 196,714]	434	0.29
Univ. of MI	Comp. Sci./IT Support	6	149,209	[100,201, 219,616]	937	0.33
Univ. of IL Urbana-Champaign	Engineering	7	139,026	[102,524, 193,422]	354	0.31
Univ. of TX - Austin	Engineering	8	138,198	[93,492, 176,053]	310	0.31
Univ. of HI at Hilo	Pharmacy	9	136,317	[115,039, 158,518]	173	0.48
Univ. of VA	Comp. Sci./IT Support	10	133,589	[104,328, 194,440]	239	0.44
VA Commonwealth Univ.	Dance	9,799	29,535	[23,235, 47,728]	33	0.17
Southeast MO State Univ.	Visual/Performing Arts	9,800	29,357	[22,629, 41,247]	46	0.27
Univ. of WI - Superior	Fine And Studio Arts	9,801	29,214	[23,506, 37,919]	41	0.21
S. OR Univ.	Fine And Studio Arts	9,802	29,097	[21,314, 40,050]	48	0.22
Univ. of Montana (The)	Drama/Theatre	9,803	28,630	[21,459, 42,445]	64	0.19
GA S. Univ.	Drama/Theatre	9,804	28,357	[20,809, 34,799]	35	0.18
Kennesaw State Univ.	Drama/Theatre	9,805	28,143	[21,871, 42,845]	49	0.14
Shenandoah Univ.	Drama/Theatre	9,806	28,126	[20,947, 37,683]	60	0.20
Central CT State Univ.	Drama/Theatre	9,807	26,002	[18,608, 37,919]	33	0.13
Morris Colg.	Health Professions	9,808	24,449	[18,677, 33,810]	37	0.25

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes.

Notes: Numbers of graduates are cumulative over three graduating cohorts for bachelor's degrees. "% of Grads in Major Ind." refers to the proportion of grads in a given credential who work in the most common industry. Because of data limitations, these refer to the parent 2-digit CIP code of the credential in question. All other statistics are calculated at the 4-digit CIP code level.

Table 5: Top 10 Fields of Study for Cohorts 2001-2003 and 2013-2015, By CIP Code Aggregation Level

Cohorts 2001-2003				Cohorts 2013-2015		
CIP Code	CIP Description	Percent Share	Rank	CIP Code	CIP Description	Percent Share
2 Digit CIP Codes						
52	Business & Related Services	29.9 %	1	52	Business & Related Services	24.5 %
13	Education	8.2 %	2	51	Health Professions	9.1 %
45	Social Sciences	6.9 %	3	42	Psychology	6.5 %
51	Health Professions	5.9 %	4	13	Education	6.5 %
42	Psychology	5.7 %	5	45	Social Sciences	6.4 %
11	Comp. Sci. & Info. Support.	5.5 %	6	9	Comm, Journalism & Related	4.9 %
9	Comm, Journalism & Related	4.7 %	7	26	Biological & Biomedical Sci.	4.5 %
30	Multi/Interdisc. Studies	3.5 %	8	14	Engineering	3.9 %
14	Engineering	3.5 %	9	11	Comp. Sci. & Info. Support.	3.9 %
26	Biological & Biomedical Sci.	3.4 %	10	50	Visual & Performing Arts	3.8 %
4 Digit CIP Codes						
52.02	Business Admin, Mgmt & Ops	9.1 %	1	52.02	Business Admin, Mgmt & Ops	8.0 %
13.12	K-12 Education	5.4 %	2	42.01	Psychology (General)	5.6 %
42.01	Psychology (General)	5.0 %	3	51.38	Nursing	5.3 %
52.13	Mgmt. Sciences & Quant. Methods	4.1 %	4	13.12	K-12 Education	4.2 %
52.03	Accounting & Related Services	3.9 %	5	26.01	Biology (General)	3.6 %
30.99	Multi/Interdisc., Other	3.2 %	6	52.03	Accounting & Related Services	3.6 %
52.08	Finance & Financial Mgmt	3.1 %	7	24.01	Liberal Arts & (General) Studies	3.0 %
51.38	Nursing	2.7 %	8	52.08	Finance & Financial Mgmt	2.4 %
26.01	Biology (General)	2.7 %	9	23.01	English Lang. & Lit. (General)	2.3 %
45.11	Sociology	2.7 %	10	43.01	Criminal Justice	2.3 %

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes.

Notes: Percent shares are based on weighted sample.

Table 6: Variance, Covariance, and Correlations of Institution and Field of Study Fixed Effects Across Graduating Cohorts

Graduation Cohort	Covariance Institution-FOS	Correlation Institution-FOS	Variance Institution	Variance FOS	Total Variance
2001–2003	0.0010	0.0629	0.0083	0.0322	0.1904
2004–2006	0.0009	0.0514	0.0089	0.0322	0.1902
2007–2009	0.0010	0.0553	0.0092	0.0376	0.2052
2010–2012	0.0009	0.0483	0.0096	0.0372	0.2052
2013–2015	0.0012	0.0624	0.0098	0.0405	0.1990

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes.

Notes: All metrics are with respect to inflation-adjusted logged average earnings.

Supplemental Appendix

Where to Study or What to Study?

Changing Horizontal Stratification in Bachelor's Degrees in the 21st Century

Contents

Supplementary Methods	A1
Comparing the Representativeness of PSEO Data to All United States Institutions of Higher Education	A1
Calculating Industry-Level Average Earnings from ACS Data	A2
Calculating Average Annual Earnings by Field of Study Adjusting for Industry Shares	A2
Assessing Changes in Field of Study Premiums Alongside Underlying Demographic Changes and Changes to Industry Flows	A3
Robustness Checks Specifications	A5
Supplemental Figures	A7
Figure A1 - Map of Included Degree-Granting Institutions Showing Unequal Coverage by State	A7
Figure A2 - Replication of Primary Analysis, Visualized by Cohort and Year of Follow-Up	A8
Figure A3 - Reanalysis of Main Text Figure 7: Institution Average Premiums, After Adjusting for Distribution of Fields of Studies, Regressed on Constant Institutional Characteristics (Average Institutional Characteristics Across All Cohorts) for Two Cohorts	A9

Figure A4 - Various Robustness Checks of the Primary Decomposition Analysis, Variably Alternating the Level of CIP Code Aggregation, Logged and Unlogged Outcomes, the Distribution Functional Form, Weights for Missingness, and the Presence of Non-Workers A10

Figure A5 - Figure 5 With Added Comparison for Expected Changes in Earnings Based on Demographic Recomposition by Field of Study. . . A11

Supplemental Tables A12

Table A1 - Fixed Effects Models Comparison - Field of Study Average Earnings Predicted By Demographics or Industry A12

Supplementary Methods

Comparing the Representativeness of PSEO Data to All United States Institutions of Higher Education

The PSEO data are based on state-, agency-, and institution-level agreements with the U.S. Census Bureau, the IRS, and the LEHD. As such, the data are not representative of all institutions of higher education that offer bachelor’s degrees. To quantify this discrepancy and adjust for it, I use an entropy balancing procedure that reweights the observed PSEO sample so that it matches the broader population of universities and colleges on a set of observable characteristics.

Using the Department of Education’s College Scorecard (also used in the main analysis), institutional attributes such as mean SAT scores, whether it is public or private, undergraduate tuition, undergraduate enrollment, graduate enrollment, and selectivity can be gleaned for the near-universe of U.S. institutions of higher education.

Formally, let the “target” group be institutions not included in the PSEO for the entire temporal span of the dataset and the “sample” group be institutions included throughout. Entropy balancing chooses nonnegative weights w_u for the included (sample) institutions to minimize the Kullback–Leibler divergence from uniform weights subject to first moment-matching constraints on observed covariates. Denoted by $\mathbf{X}_u$ in equation 1 is the vector of institution-level characteristics:

$$\mathbf{X}_u = [\text{Private}_u, N_u^{\text{gr}}, \log N_u^{\text{gr}}, N_u^{\text{ug}}, \log N_u^{\text{ug}}, \log(\text{Tuition}_u), \text{SAT}_u, \text{AdmRate}_u]^\top \quad (1)$$

Equation 2 shows the optimization problem.

$$\min_{\{w_u \geq 0\}} \sum_{u \in \text{Sample}} w_u \log w_u \quad \text{s.t.} \quad \sum_{u \in \text{Sample}} w_u \mathbf{X}_u = \frac{1}{N_{\text{Target}}} \sum_{u' \in \text{Target}} \mathbf{X}_{u'}, \quad \sum_{u \in \text{Sample}} w_u = 1 \quad (2)$$

Intuitively, this assigns more weight to included institutions whose characteristics are underrepresented relative to the target and less weight to those that are overrepresented, while preserving as much of the original distribution as possible.

I limit extreme weights to their values at the 2.5th and 97.5th percentiles to mitigate the influence of outliers. The resulting weights are applied to the included institutions in all reweighted summaries and figures. The effectiveness of the reweighting (i.e., balance on the covariates listed above before and after weighting) is displayed in the main text in Table 2.

Calculating Industry-Level Average Earnings from ACS Data

Since graduation year is not observed in the ACS, I assume that all graduates were on average 23-years-old upon graduation, and I used the annual earnings of all workers working full time who are aged 27-29 at the time of the survey. While not a perfect solution, the methods described below will show that it only matters for calculating annual earnings relative to each other since the primary datasource remains the PSEO-based distributions of annual earnings by field of study.

Calculating Average Annual Earnings by Field of Study Adjusting for Industry Shares

These average annual earnings by field of study and industry net of each other $\alpha_{f,y}$ and $\beta_{ind,y}$, respectively, in a given year (y) can be computed based on share of flows from field of study to industry as a proportion of total graduating students $S_{f,ind,y}$. We may assume the following additive decomposition model (equations 3 and 4):

$$\omega_{f,y} = \alpha_{f,y} + \sum_{ind} S_{f,ind,y} \cdot \beta_{ind,y} \quad (3)$$

$$\omega_{ind,y} = \beta_{ind,y} + \sum_f S_{f,ind,y} \cdot \alpha_{f,y} \quad (4)$$

Where $\omega_{f,y}$ is the observed average annual earnings for a field of study in a given year, known from the LEHD/PSEO data, and $\omega_{ind,y}$ is the average annual earnings in an industry in a given year for all college graduates, known from the ACS data. To solve the above system of equations, I must simply add a constraint that the mean industry effects are mean zero:

$$\sum_{ind,y} \beta_{ind,y} = 0 \quad (5)$$

This system of three equations (equations 3-5) may then be solved using least squares, performed separately for each year. With the final estimates of $\hat{\alpha}_{f,y}$ and $\hat{\beta}_{ind,y}$, which correspond to the industry-adjusted field of study-specific average wages and the field of study-adjusted industry-specific average wages respectively, it is possible to calculate average expected annual earnings by field of study and industry combination.

Using the estimates of $\hat{\alpha}_{f,y}$ and $\hat{\beta}_{ind,y}$ from above combined with flows data, the predicted aggregate average annual earnings for a given field of study in a particular year ($\omega'_{f,y}$) can be compared to their observed values ($\bar{\omega}_{f,y}$), giving a sense of the extent to which observed shifts in annual earnings are due to changing flows to industry and industry-level expansion and contraction. It is worth noting at this moment that this analysis relies on the additivity of industry effects and field of study effects and that there be no interactive effects between the two. Otherwise put, a student with a computer science degree and a liberal arts degree, both working in the same industry of “Information” would earn salaries commensurate with the additive effects of their fields of study and industries. This would not allow for any synergistic effects for computer science majors, given they are working in an industry that is especially proximate to their field of study. While this is a strong assumption, it once again makes my estimates an underestimate of the total effect of industry in the following analysis since interactive effects are disregarded. Further, in the following analysis, I explicitly quantify this residual effect for comparison and interpretation.

Assessing Changes in Field of Study Premiums Alongside Underlying Demographic Changes and Changes to Industry Flows

It is not mutually incompatible that changing field of study average wages may reflect changing industry destinations and industry-level average wages and that fields of study

may be changing demographically, which in turn changes the prestige or desirability of a given field of study. Indeed, these two processes may be happening simultaneously and/or because of each other without being necessarily competing explanations. Nevertheless, comparing how parsimonious each is in explaining changing field of study-level premiums is a natural first step in understanding which question is more first-order.

In order to compare how well each source of variation predicts changes in average wages in the time period examined, I pursue a fixed effects model that fully accounts for between-unit heterogeneity and adds within-unit continuous regressors. I then examine the within-unit R^2 in order to understand how well each does. As with most primary analyses, this is performed on wages for recipients of bachelor's degrees, five years after they graduated. For each field of study-specific average wage $\bar{w}_f$, observed separately for each graduation year y , X years after graduation, I regress it on field of study-specific fixed effects F_f . To test the predictive power of demographic change, I add a vector of field of study-specific demographic variables $\xi_{f,y}$, which includes the percentage in each field of study that is male or female and either white, black, asian, and hispanic (the intersection thereof). This is eight continuous variables in total. To test the predictive power of industry flows, I simply add the expected average income based on the industry-weighted average of industry-specific average wages, as explained in the main text. This is a singular variable.

As this is a fixed effects model, the static characteristics of each field of study will be captured by the field of study fixed effect. Baseline values of either expected wages based on industry composition or demographic characteristics will be “netted out,” so the only source of variation will be within-unit changes in these continuous indicators. Thus, I will use the within-unit R^2 to understand the predictive power of $\xi_{f,y}$ and $\ln(\bar{w}_{f,Y}^*)$, alternatively. A value of 1 would indicate 100% explanatory power of either to explain wage changes over the period observed, as compared to 0, which would indicate no predictive power. Using these methods, I can also create an alternative version of the main text's Figure 5, which shows predicted changes in field of study average wages based on demographic changes.

$$\ln(\bar{\omega}_f|Y = y + X) = F_f + \xi_{f,y} + \epsilon_{f,y} \quad (6)$$

$$\ln(\bar{\omega}_f|Y = y + X) = F_f + \ln(\bar{\omega}_{f,Y}^*) + \epsilon_{f,Y} \quad (7)$$

Regression estimates are shown in in Appendix Table A1, and an alternative version of Figure 5 is shown in Appendix Figure A5, below.

Robustness Checks Specifications

Appendix Figure A4 displays eight robustness checks. Below, each is detailed in the order that it appears in the figure.

- **2 Digit CIP Codes + Log-Normal Distribution** – I replace 4 digit CIP Codes (N = 267) with parent 2 digit CIP Codes (N = 36). As in the primary analysis, the outcome (earnings) is logged.
- **2 Digit CIP Codes Without Logged Outcomes** – I replace 4 digit CIP Codes (N = 267) with parent 2 digit CIP Codes (N = 36). The outcome (earnings) is untransformed.
- **2 Digit CIP Codes WWithout Logged Outcomes** – I replace 4 digit CIP Codes (N = 267) with parent 2 digit CIP Codes (N = 36). The outcome (earnings) is untransformed. I include non-workers as earners with an income of \$0. The number of workers not employed is available at the credential-cohort level for 2 digit CIP Codes, though it is not available at the 4 digit Level.
- **4 Digit CIP Codes + Log-Normal Distribution** – This is the primary analysis, where the outcome is log-normalized, and 4 digit CIP Codes are used.
- **4 Digit CIP Codes + Pareto Distribution** – I replace the assumed log-normal distribution within each credential with a piecewise log-normal/Pareto distribution. This distribution is identical for all workers below the 98th percentile. For those above the 98th percentile, I replace the assumed log-normal distribution within each credential with draws from a Pareto distribution with an α value of 2.
- **4 Digit CIP Codes + Reordered Fixed Effects** – This specification is identical to the primary analysis but reverses the order of inclusion of fixed effects, with institutional fixed effects added before those for field of study. In practice, this approach necessarily attributes a larger share of earnings variation to institutions, since differences by field are only measured after controlling for where students

studied. In this scenario, the institutional effects also capture differences that may in fact be attributable to variation in fields of study.

- **4 Digit CIP Codes + Unweighted** – This analysis uses the same specification as the primary model but does not weight use the weights calculated to make the sample more closely match the target population of all U.S. schools offering 4-year degrees.
- **Semi-Aggregated (N = 64) 4-Digit CIP Codes + Log-Normal Distribution** – This analysis uses the same specification as the primary model but replaces the 267 CIP codes with a more aggregated set of 64 CIP codes. As opposed to using 2 digit CIP codes, this uses the 30 largest 4 digit CIP codes alongside “remainders.” For instance, CIP code 13.12 or “K-12 Education” is used, and a new “remainder” CIP code is created for all CIP codes that begin with 13 but are not 13.12. For 2 digit parent CIP codes with multiple children 4 digit CIP codes that are in the top 30 CIP codes, all get separate entries before calculating the remainder. 2 digit CIP codes with no children 4 digit CIP codes constituting the top 30 are still included. This condition is created to show that it is not merely mechanistic that fields of study capture so much variation compared to institutions, as much of the result can be replicated with a restricted set of CIP codes.
- **All Data (Unbalanced Panel)** – This analysis uses the same specification as the primary model but includes all possible data, including those from schools who enter the PSEO data system after the beginning of the dataset, thereby forming an “unbalanced panel.” Data are unweighted.

Supplemental Figures

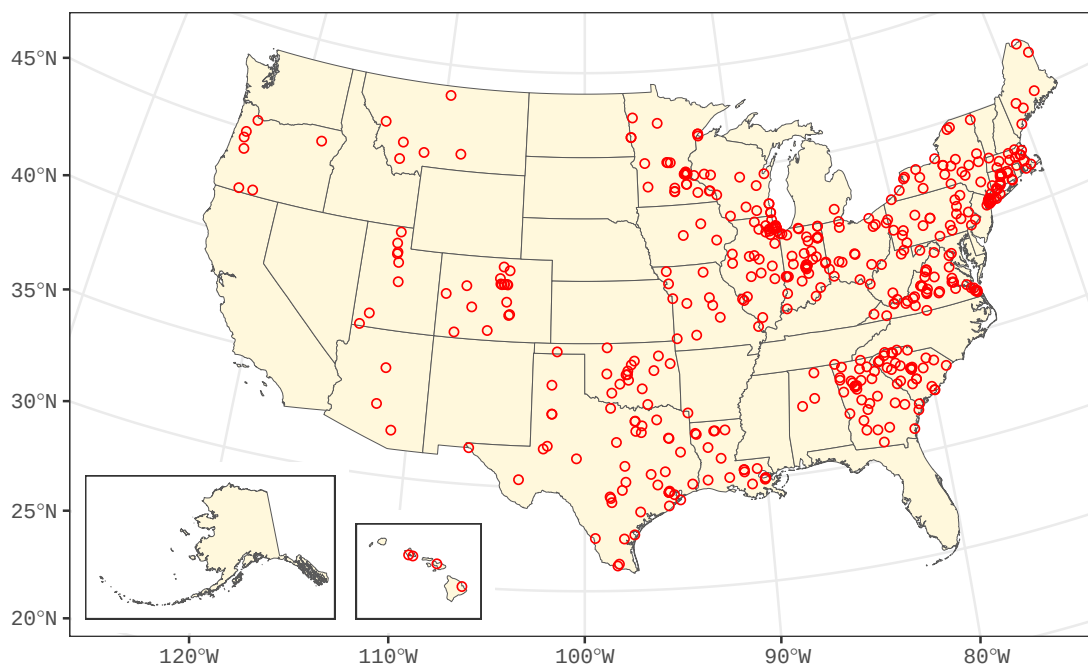


Figure A1. Map of Included Degree-Granting Institutions Showing Unequal Coverage by State

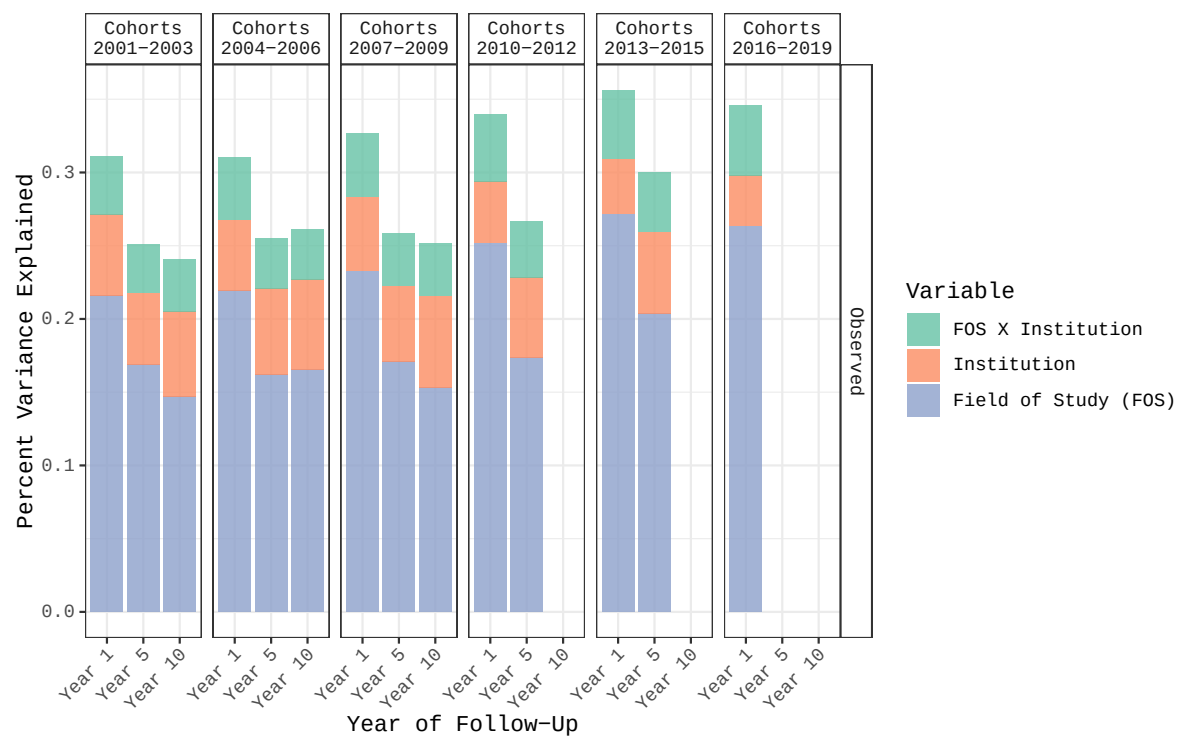


Figure A2. Replication of Primary Analysis, Visualized by Cohort and Year of Follow-Up

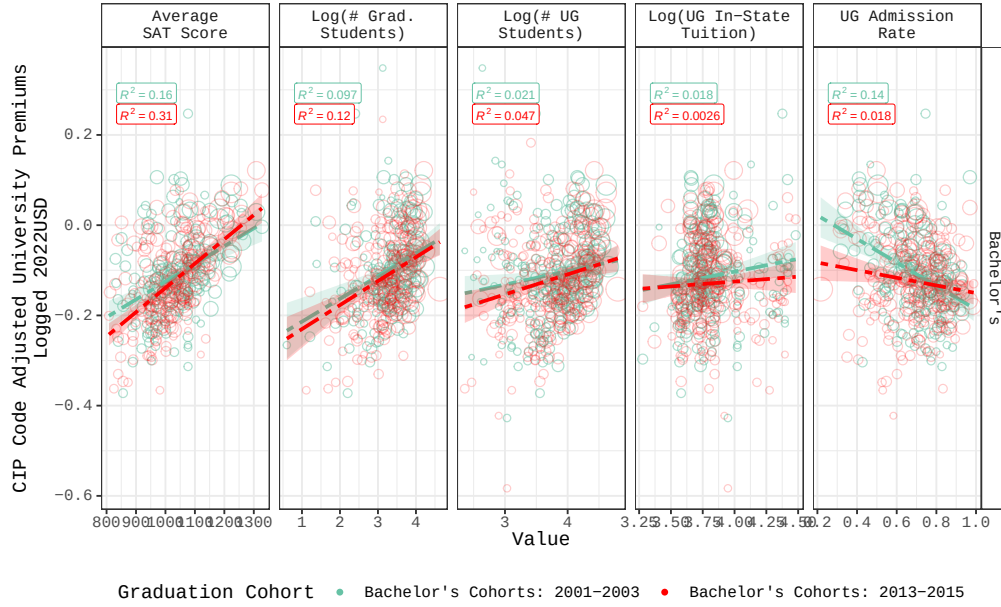


Figure A3. Reanalysis of Main Text Figure 7: Institution Average Premiums, After Adjusting for Distribution of Fields of Studies, Regressed on Constant Institutional Characteristics (Average Institutional Characteristics Across All Cohorts) for Two Cohorts

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes; The U.S. Department of Education's College Scorecard.

Notes: Patterns for the relationship between institutional premiums and institutional characteristics are generally unchanged from the primary analysis, with slight variations in magnitude.

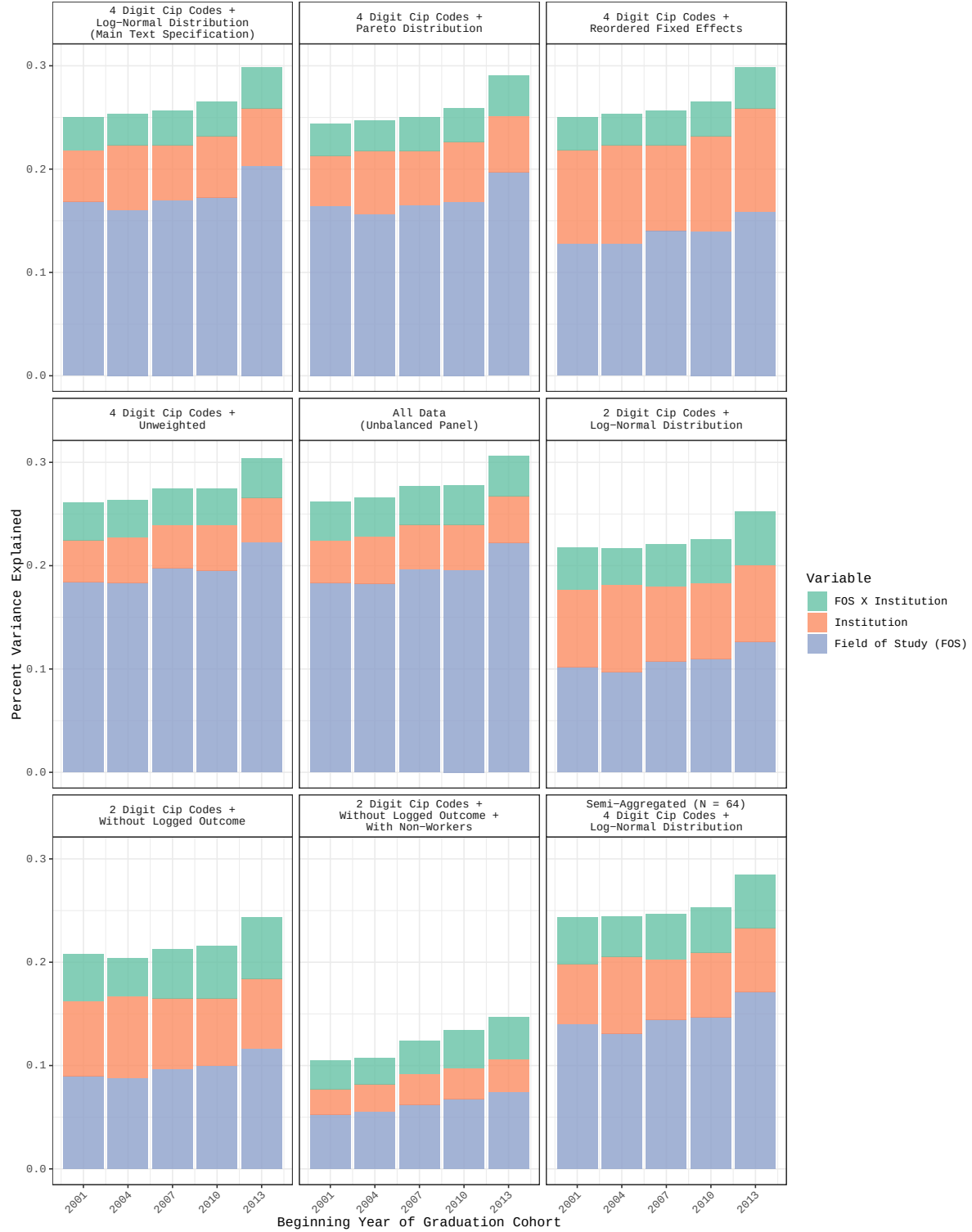


Figure A4. Various Robustness Checks of the Primary Decomposition Analysis, Variably Alternating the Level of CIP Code Aggregation, Logged and Unlogged Outcomes, the Distribution Functional Form, Weights for Missingness, and the Presence of Non-Workers

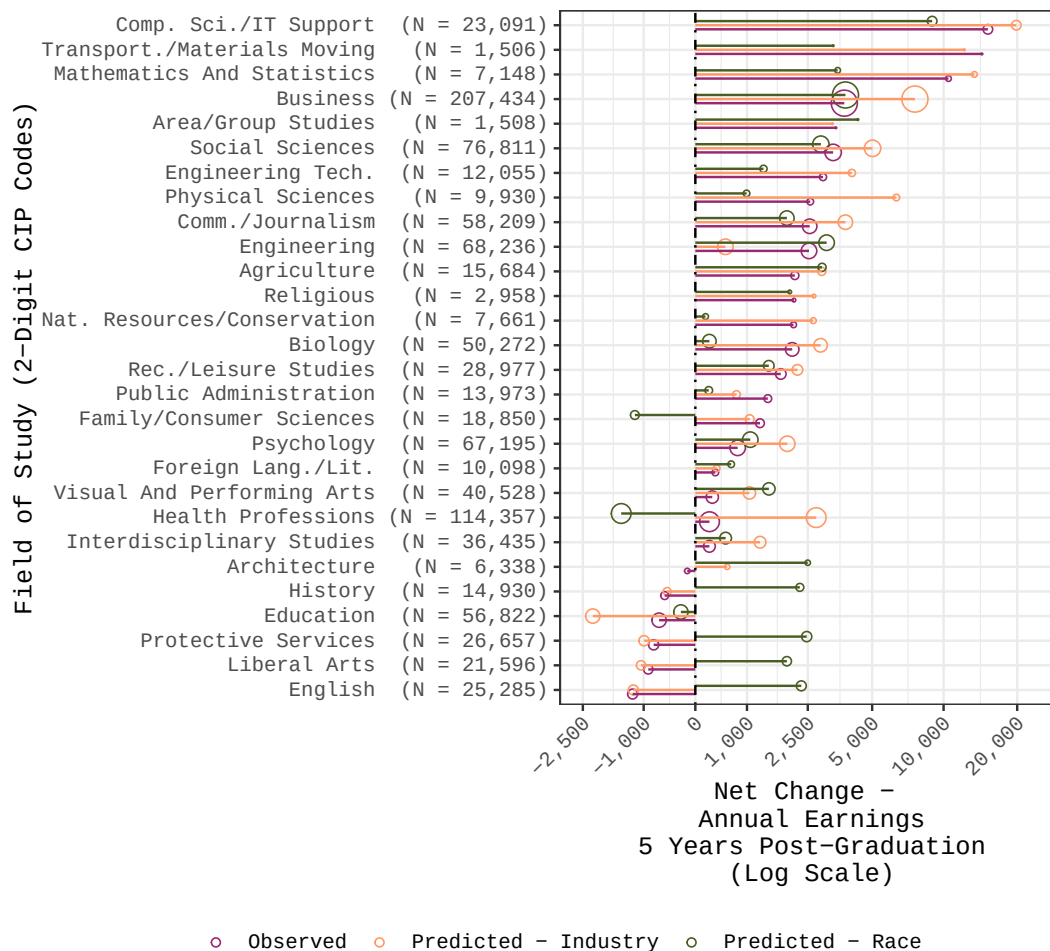


Figure A5. Figure 5 With Added Comparison for Expected Changes in Earnings Based on Demographic Recomposition by Field of Study.

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes, American Community Survey, National Center for Education Statistics, The College Scorecard.

Notes: Only fields of study with at least 1,000 graduates per year (3,000 per cohort span) are shown. Size of circles is correlated with the number of students graduating in each cohort. “N = ” for each cohort refers to their absolute sizes for graduating cohorts 2013–2015.

Supplemental Tables

Table A1. Fixed Effects Models Comparison - Field of Study Average Earnings Predicted By Demographics or Industry

Model:	Log(Average Earnings)	
	(1)	(2)
<i>Variables</i>		
Pred. Earnings Based on Industry (Logged)	0.7980*** (0.0552)	
White Men		-0.2922 (0.3656)
White Women		0.0472 (0.2120)
Black Non-Hispanic Women		-0.9098 (0.9289)
Black Non-Hispanic Men		-0.0610 (1.4813)
Asian Men		1.7255* (0.8260)
Asian Women		-2.9202** (1.0579)
Hispanic Women		-0.1739 (0.5425)
Hispanic Men		0.6782** (0.2433)
<i>Fixed-effects</i>		
2-Digit CIP Code (32)	Yes	Yes
Graduation Cohort (5)	Yes	Yes
<i>Fit statistics</i>		
Observations	160	160
R ²	0.99623	0.98104
Within R ²	0.85615	0.27669
<i>Clustered (2-Digit CIP Code) standard-errors in parentheses</i>		
<i>Signif. Codes: ***: 0.001, **: 0.01, *: 0.05, +: 0.1</i>		

Data source: Longitudinal Employer-Household Dynamics Post-Secondary Education Outcomes, American Community Survey, National Center for Education Statistics, The College Scorecard.

Notes: Demographics are the percent of each field of study that belong to each category. They do not sum to one, as there is a remainder that belong to one or more races and American Indian and Alaska Natives.